

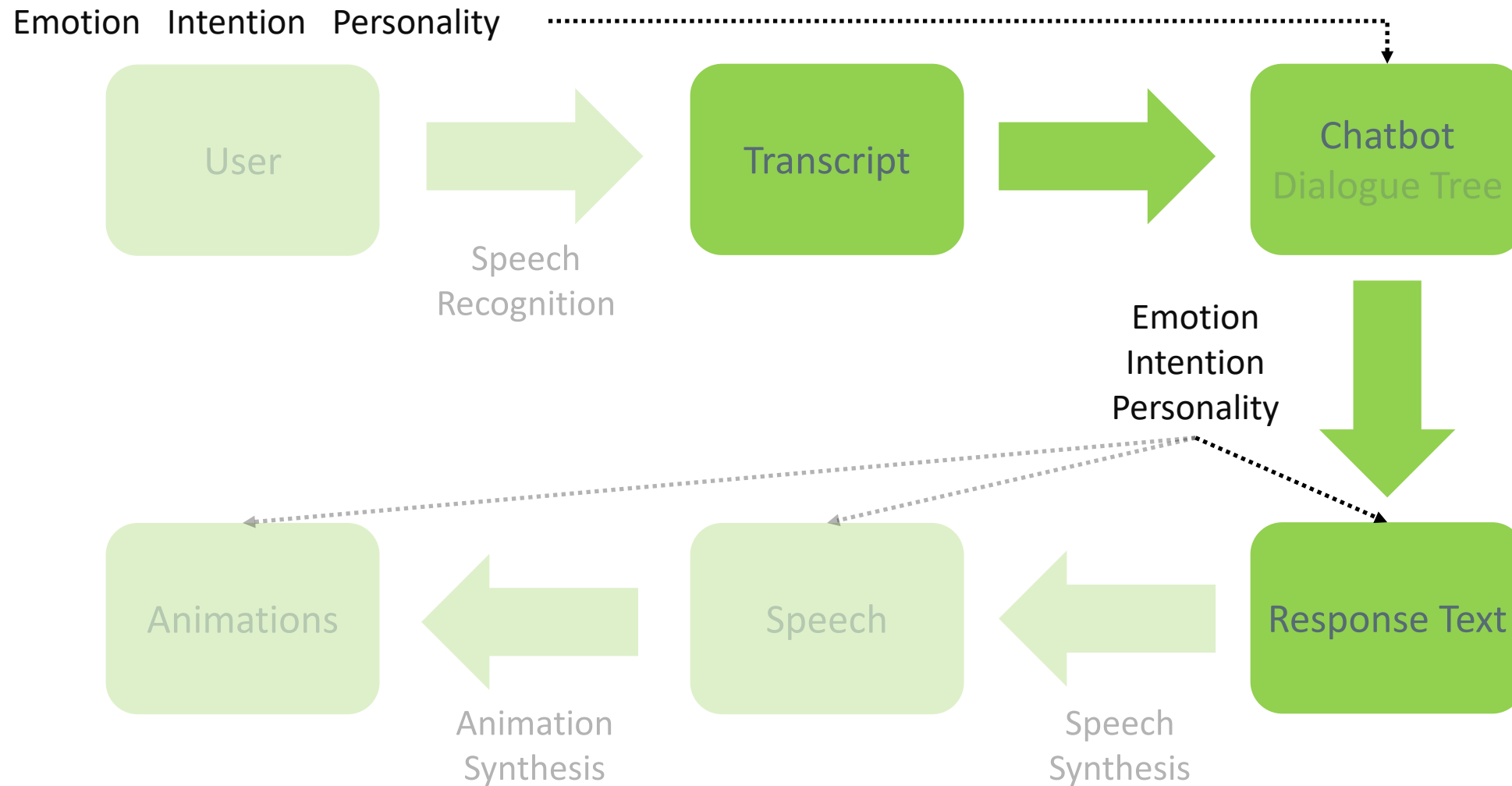


Artificial Intelligence for Digital Characters

Chatbot III: Multimodality and Affective Intelligence

Xiang Zhang, ETH Zurich

Conversational Digital Character

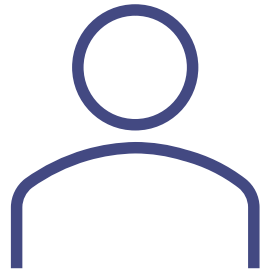


Multimodality

Emotion Recognition
Personality Modeling

Input Modalities
Output Modalities
User State Modeling

So far ...



User

“Text”



Chatbot



Audio



Video

What we want ...

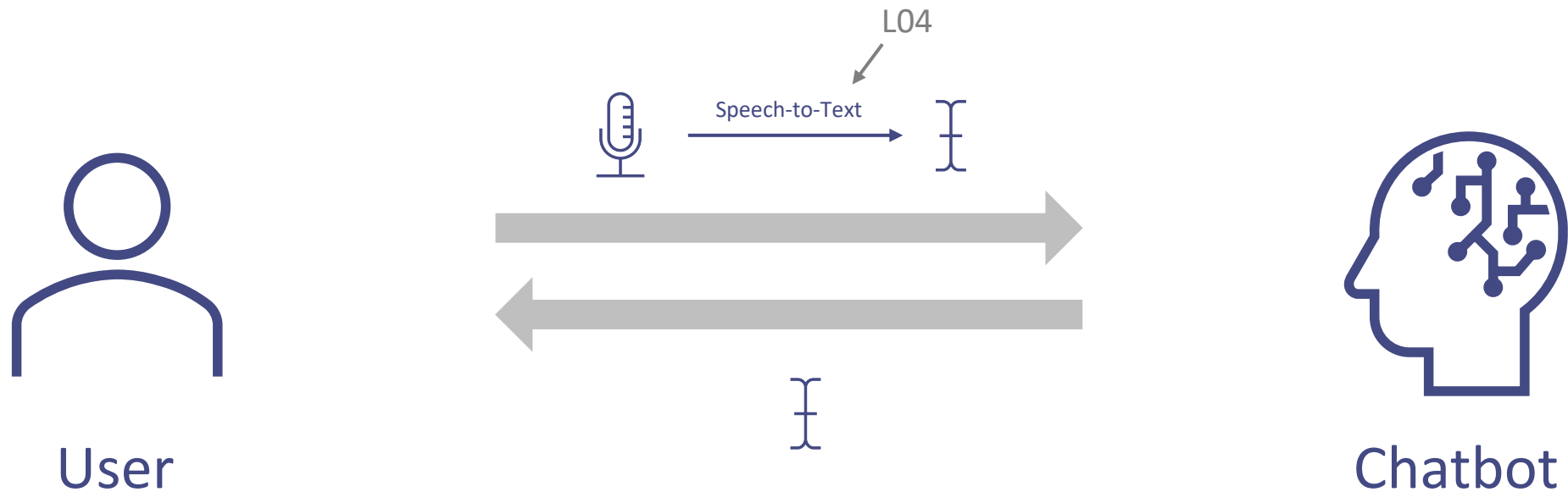


Speech

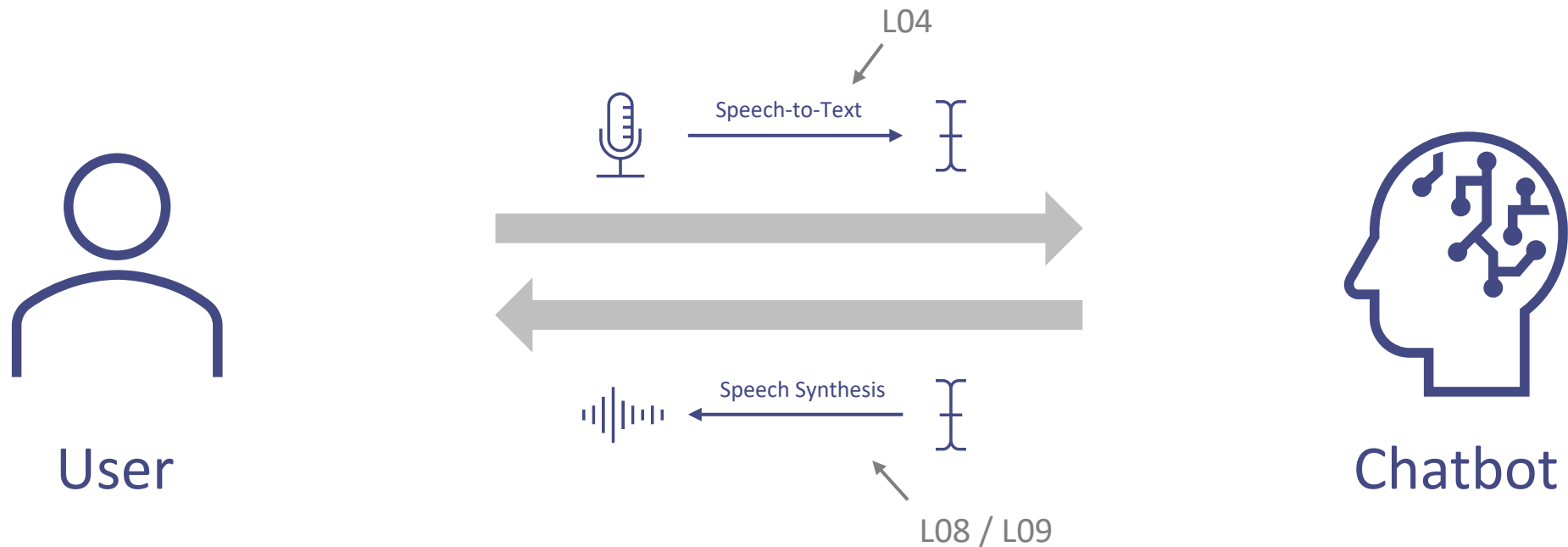


Embodiment

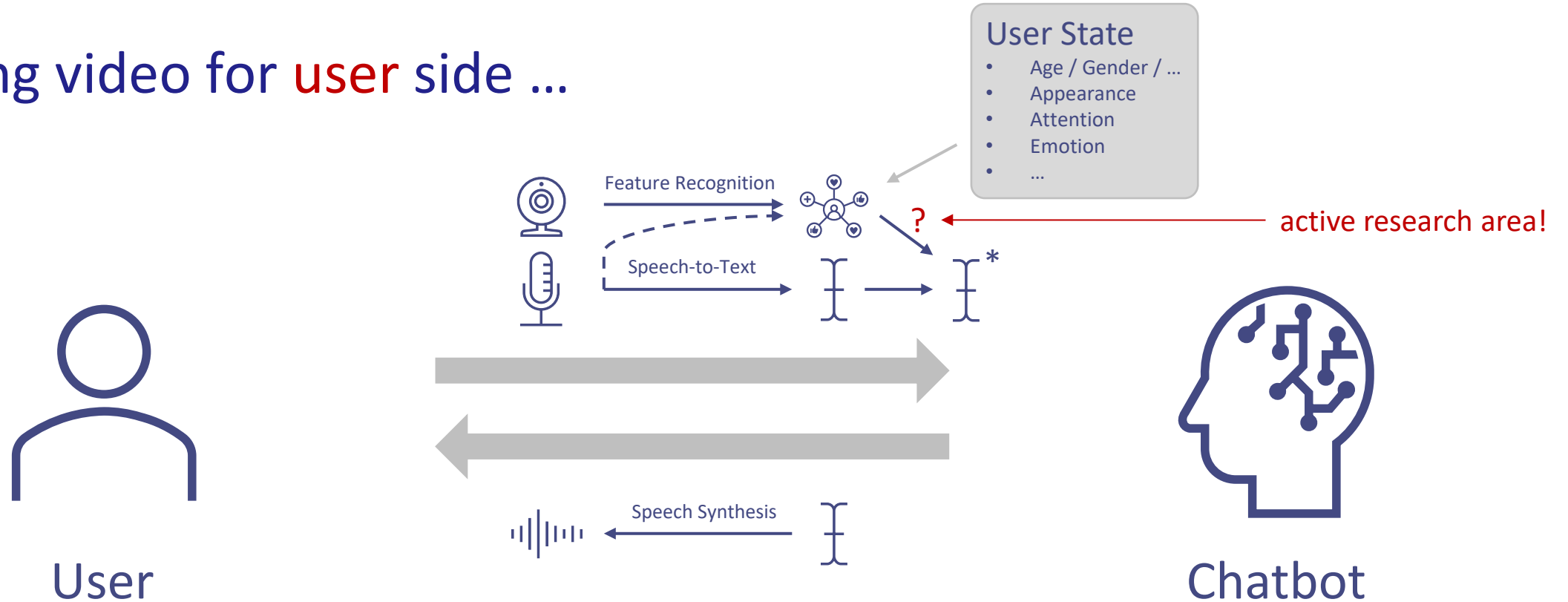
Adding audio for **user** side ...



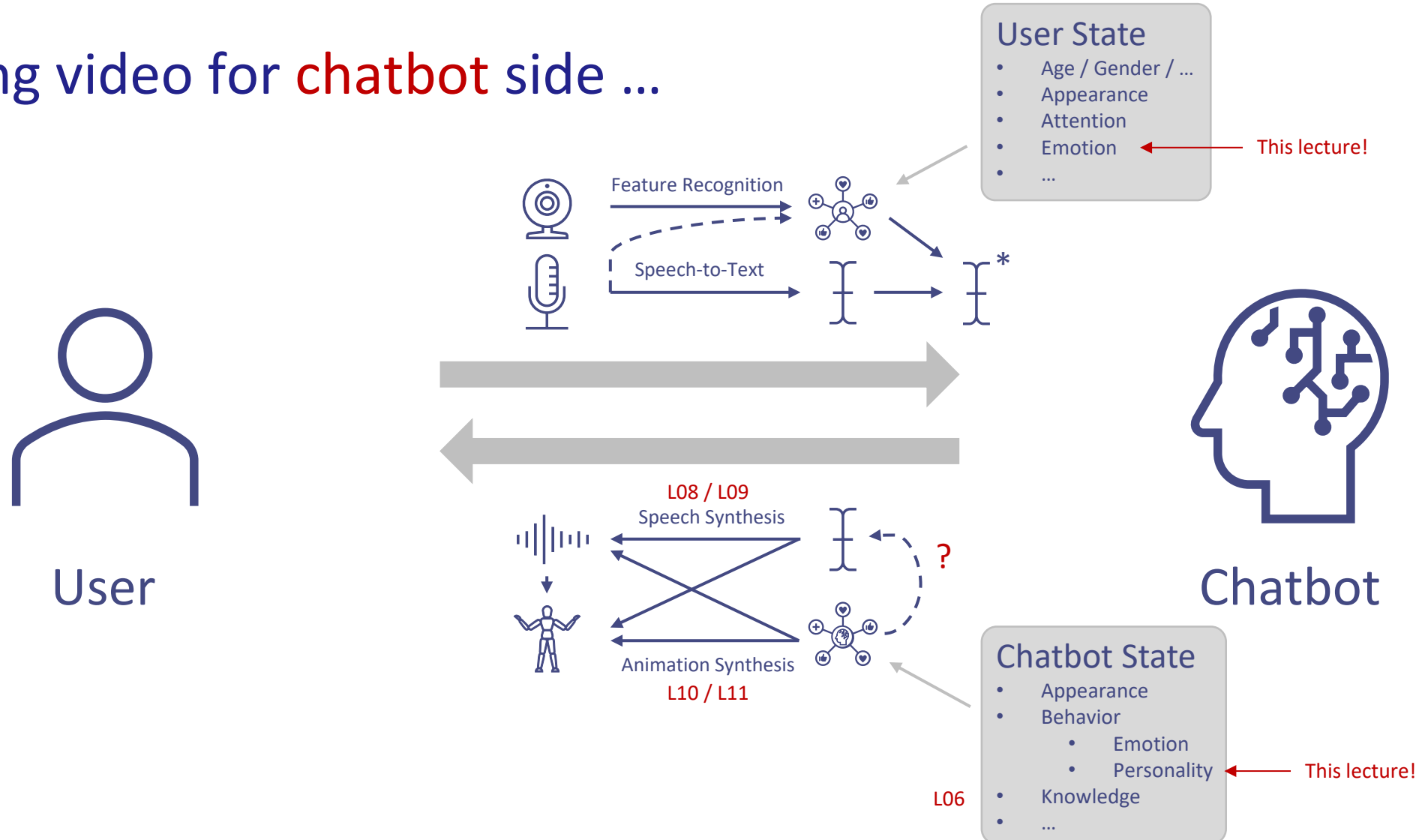
Adding audio for **chatbot** side ...



Adding video for **user** side ...



Adding video for **chatbot** side ...

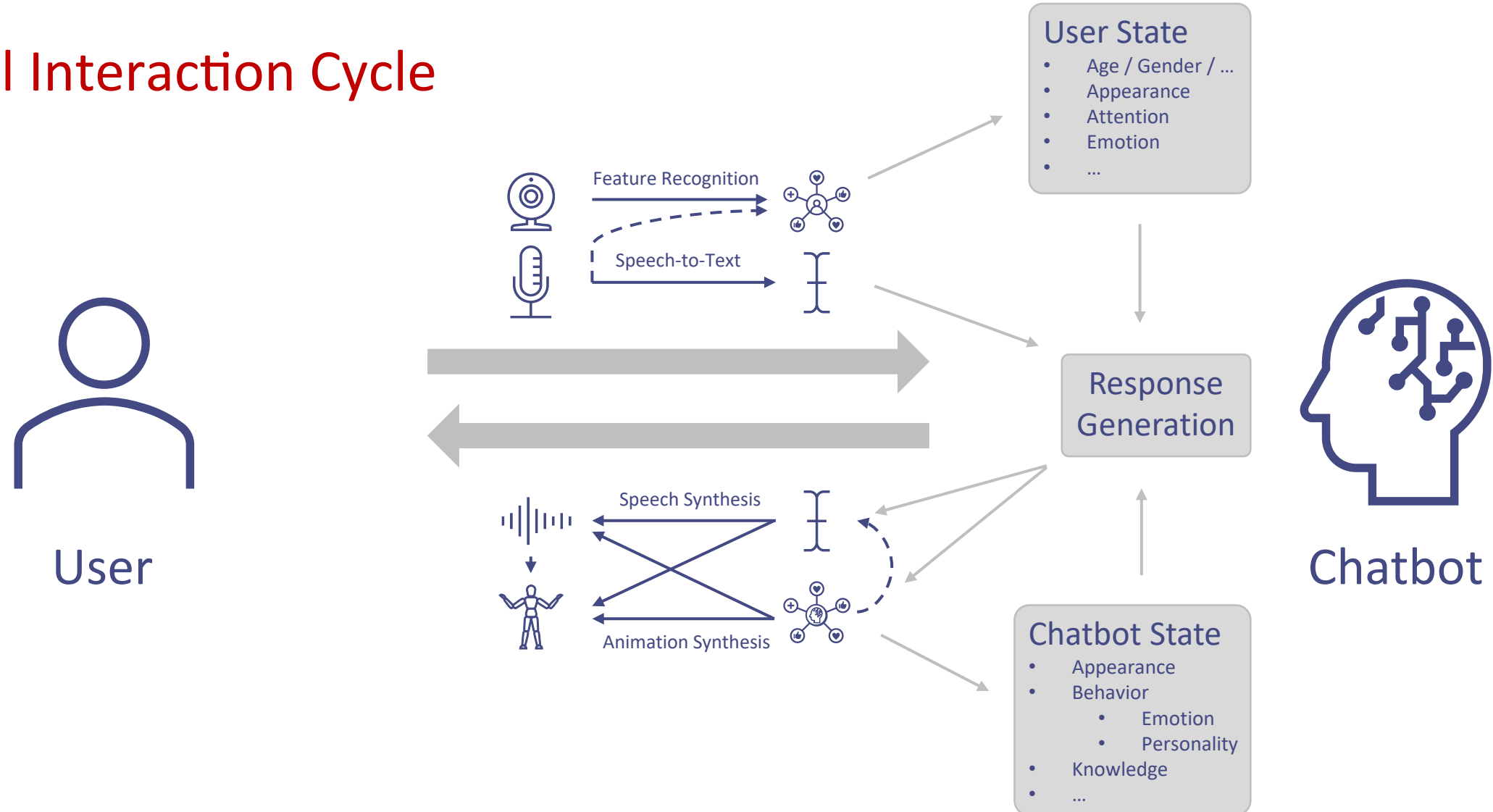


Digital Einstein

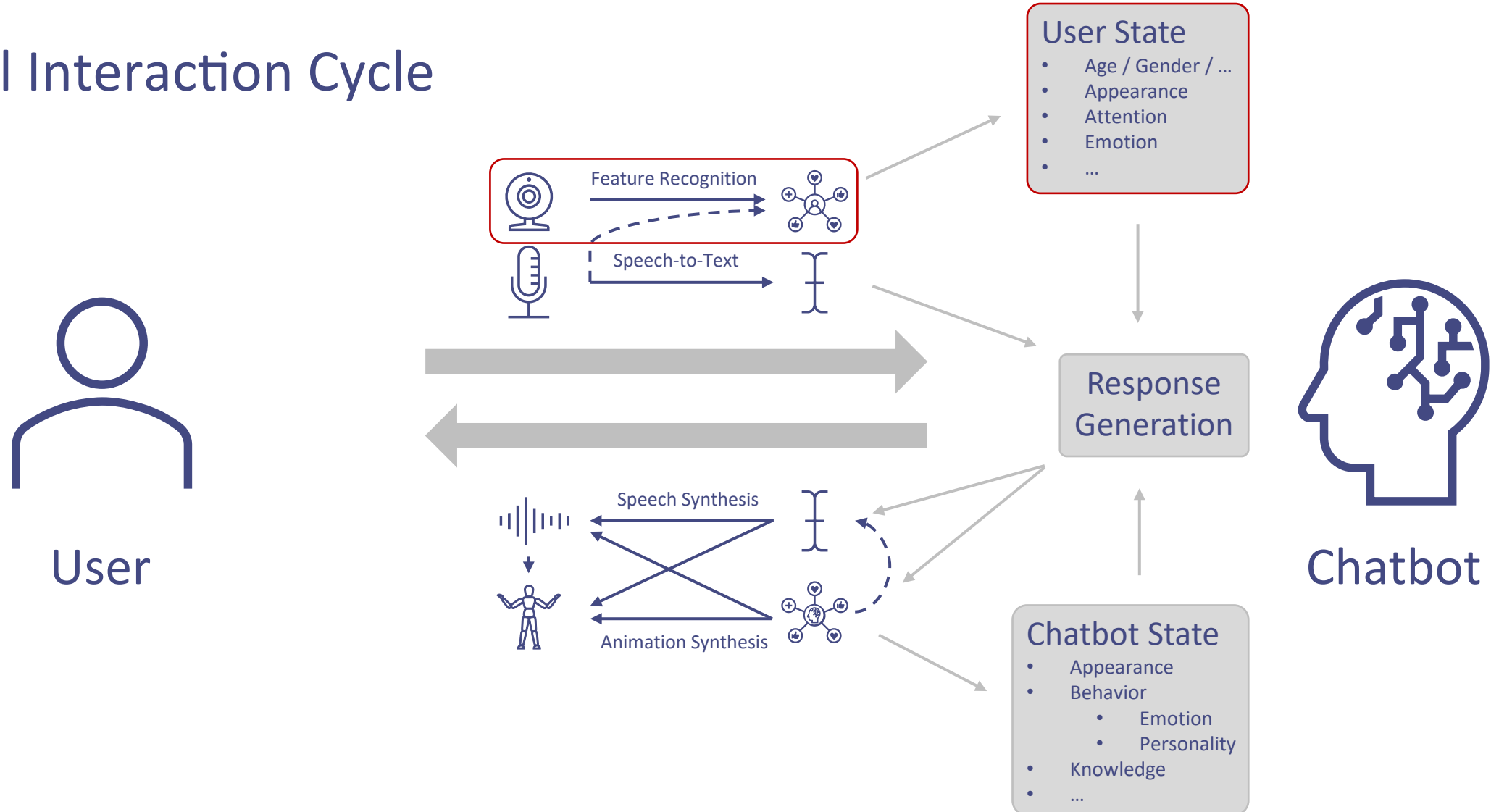


-  **User**
-  **Camera**
-  **Microphone**
-  **Chatbot**
-  **Speakers**
-  **Display**

Full Interaction Cycle



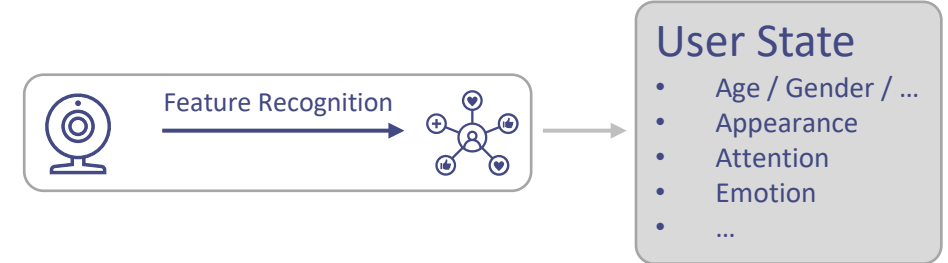
Full Interaction Cycle



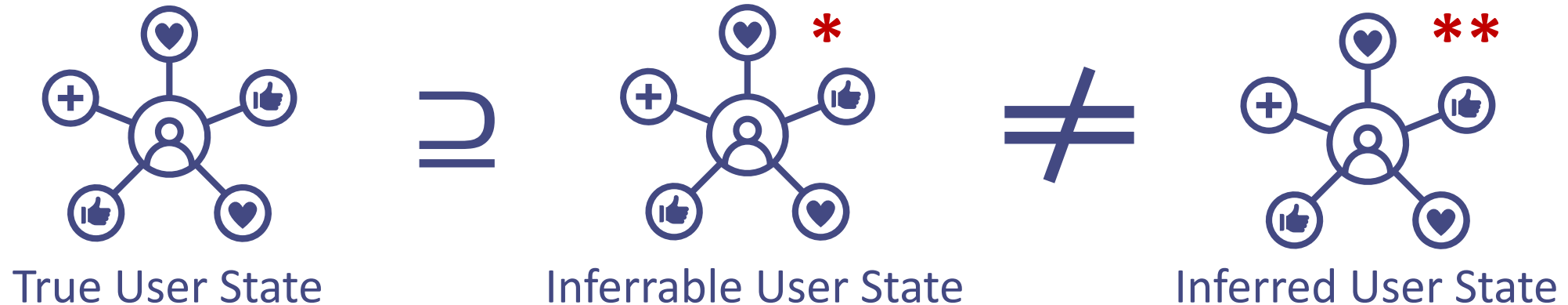
What is the «User State» ?

User State.

Anything about the user that could be of relevance for the response generation.



← This depends on the application scenario!



Appearance

Visual Features

Convolution-based networks

- VGG
- EfficientNet
- ...

Transformer-based networks

- ViT (vision transformers)
- Swin transformers
- ...

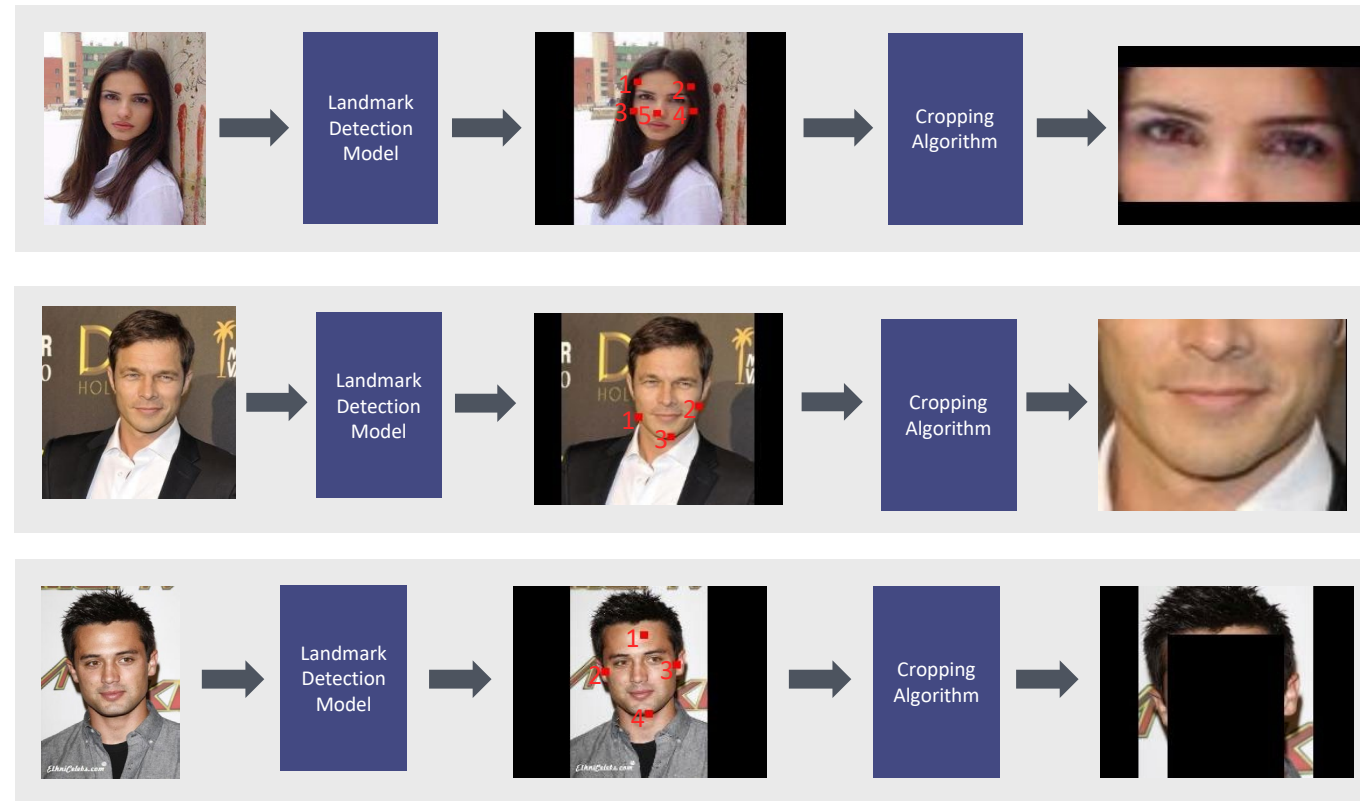


Feature Recognition



User State

- Age / Gender / ...
- Appearance
- Attention
- Emotion
- ...



Glasses?

Beard?

Hair Color?

[Credits: Yves Bischofberger]

Attention

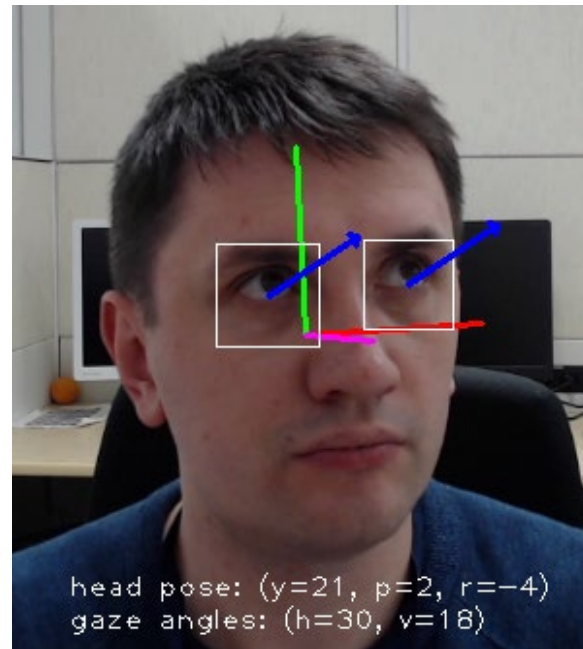
Visual Features

Convolution-based networks

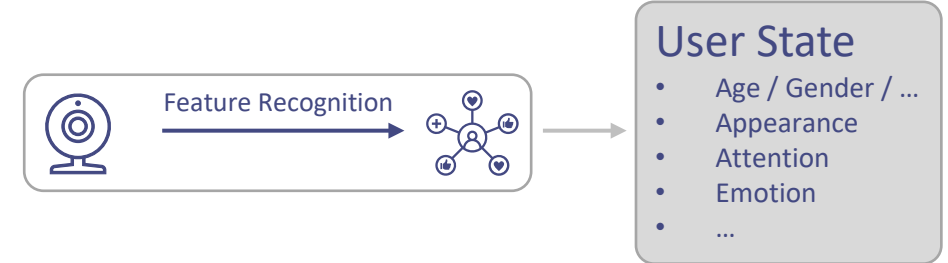
- VGG
- EfficientNet
- ...

Transformer-based networks

- ViT (vision transformers)
- Swin transformers
- ...



[Credits: OpenVINO.ai]



Emotion



Feature Recognition



User State

- Age / Gender / ...
- Appearance
- Attention
- Emotion
- ...

Visual Features

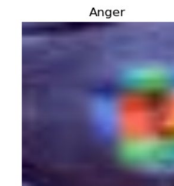
Convolution-based networks

- VGG
- EfficientNet
- ...

Transformer-based networks

- ViT (vision transformers)
- Swin transformers
- ...

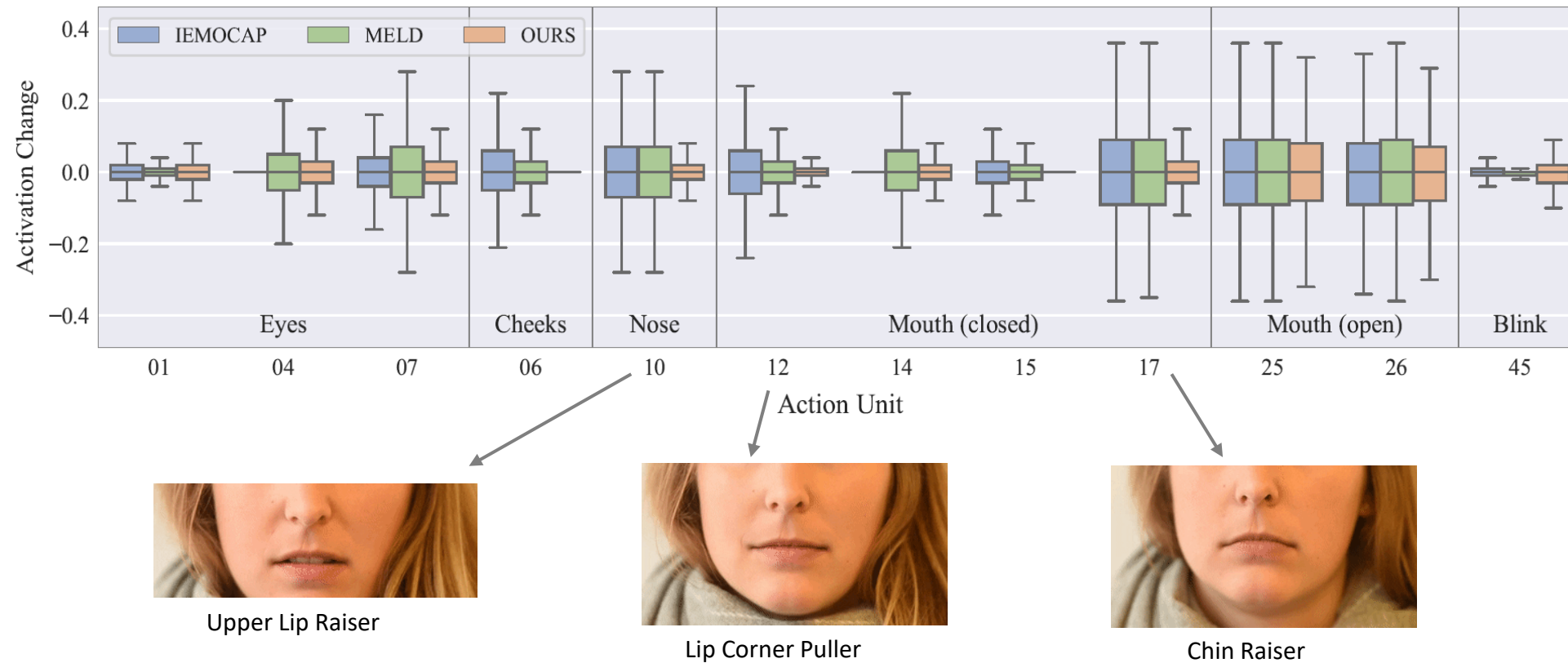
Facial Action Coding System (FACS)



How many faces are in this image?

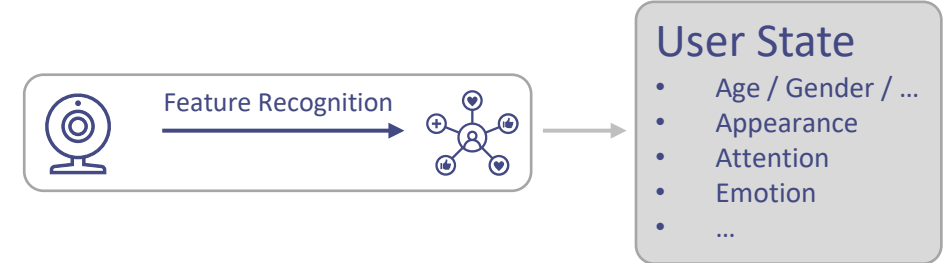
Very robust! So, problem solved?

Human-to-Human (Acted) vs. Human-to-Agent (Kovačević et al., 2024a)



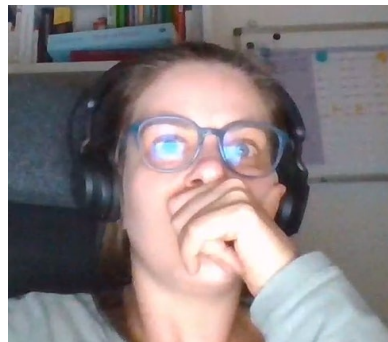
Images taken from the FACS at: <https://imotions.com/blog/learning/research-fundamentals/facial-action-coding-system/>

Emotion: Challenges



Subtlety

happy or neutral?
both plausible



Occlusions / Reflections

mouth is covered
eye gaze hardly visible



Conversation

surprised or speaking?
single frames look unnatural



Calibration / Normalization

different lighting
different mean expression

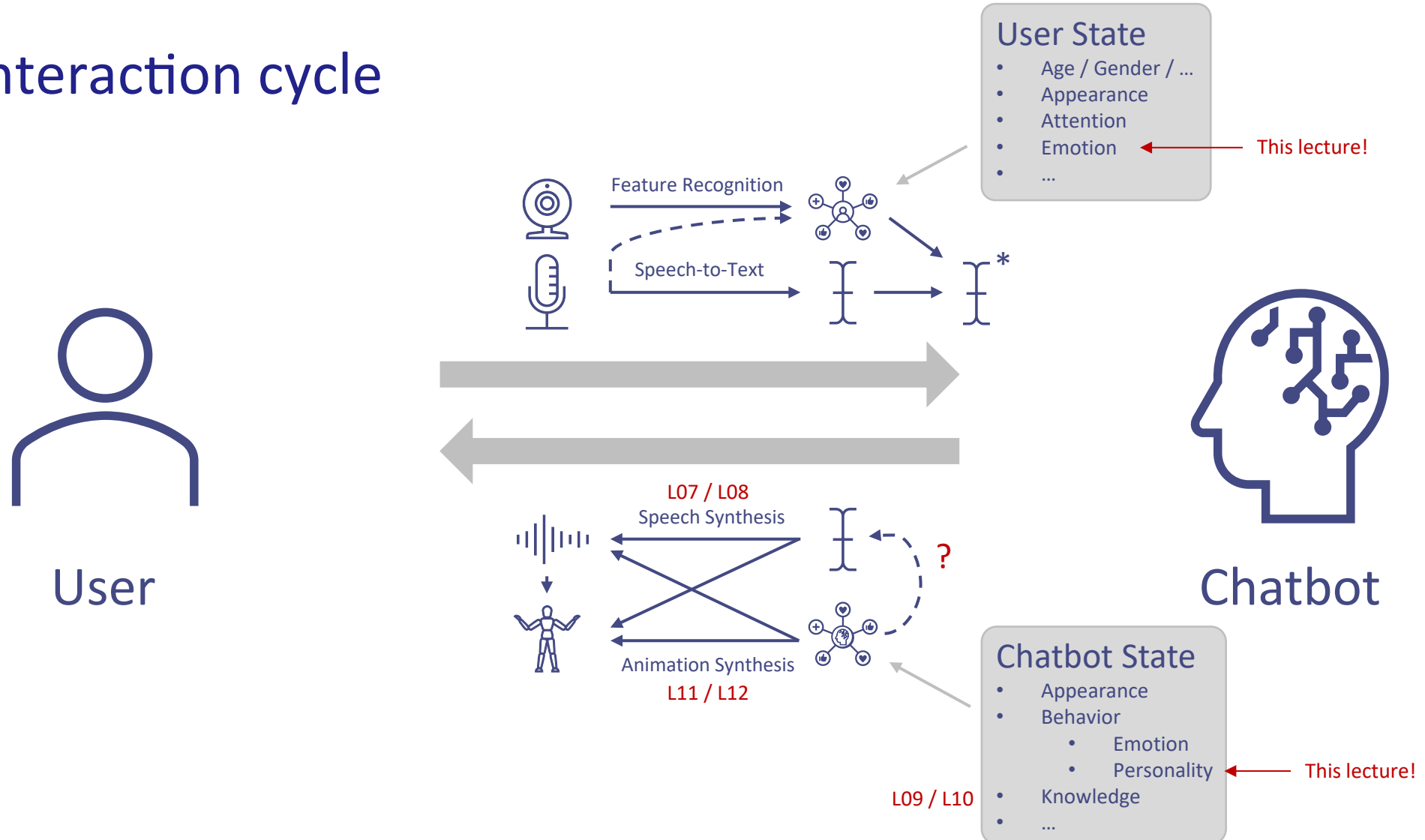
Multimodality is hard!

- Many different input/output sources
- Preprocessing needed on the fly
- Modules often work in parallel
- Real-world scenarios pose new challenges

What we didn't talk about...

- What about hardware? Constraints?
 - What about efficiency?
- *Whole separate story!*

Full interaction cycle



Emotion Recognition

Multimodality
Personality Modeling

Emotion Theory
Recognition Models
Feature Fusion
Personalization

What are emotions?

*Emotions are mental states brought on by neurophysiological changes. There is **no** scientific consensus on a definition.*

Emotion Taxonomies

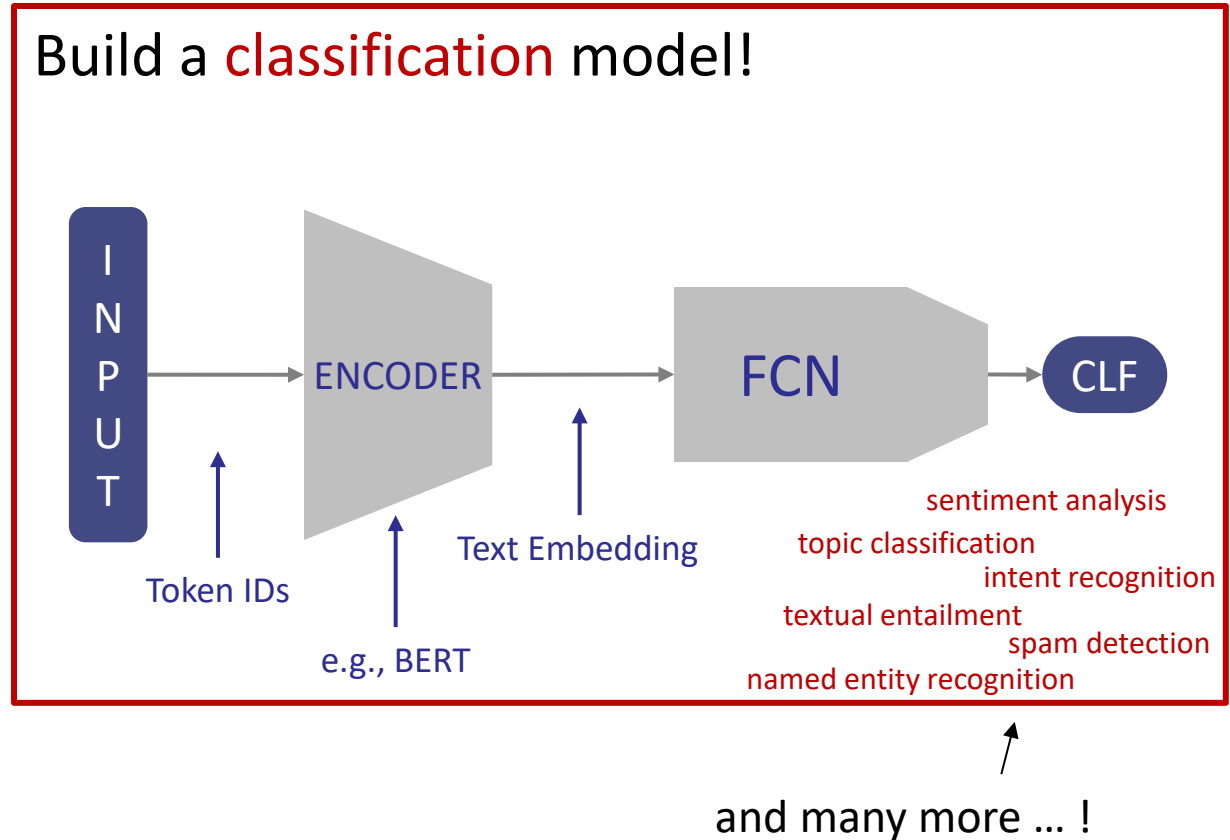
VAD Taxonomy
Dimensional Model
Russell and Mehrabian

Basic Emotion Taxonomy
Categorical Model
Ekman and Friesen

Remember Encoders and Embeddings?

- Interpret embedding as a feature vector.
- Train a classification head on these features.
- Many pretrained classification heads exist.
- Numerous tasks available!
- For example, see HuggingFace.

But this is *text*! What about audio / video ?



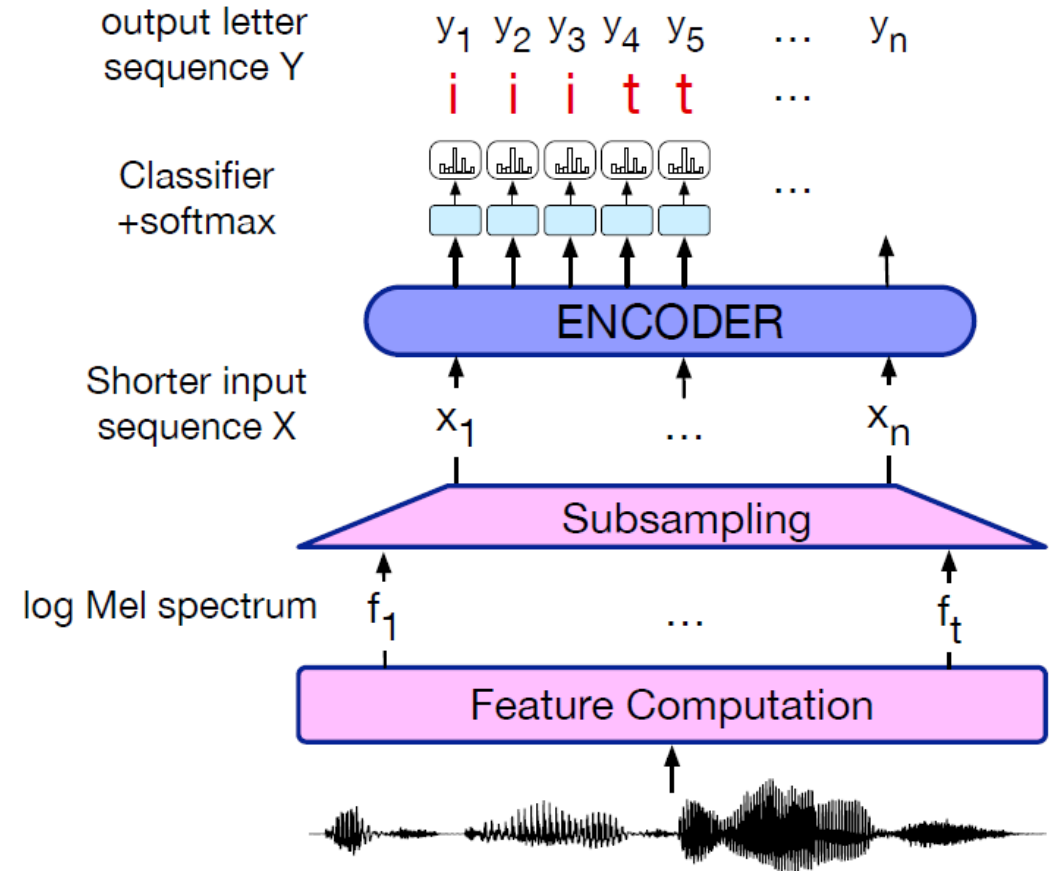
Audio Embeddings

From L04: Speech Recognition

- Audio is also represented as a list of numbers
- Can embed this representation similarly to text
- Use a classification head for various prediction tasks

What features are relevant for emotion?

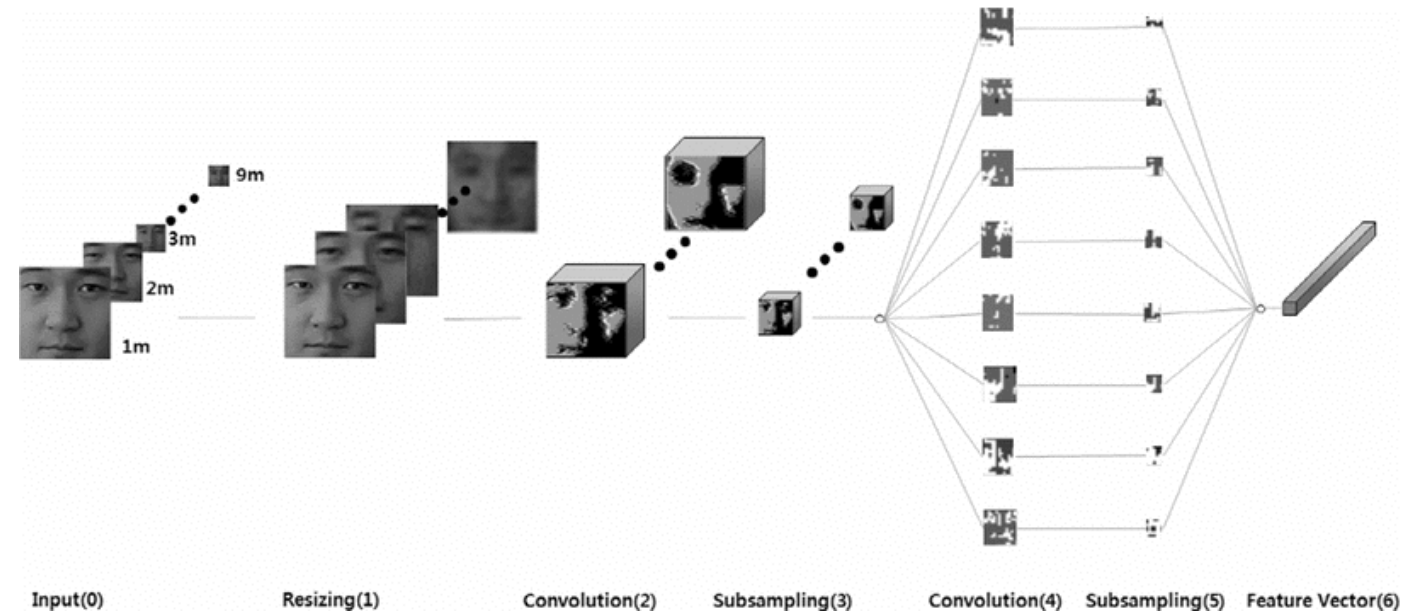
→ pitch, rhythm, intonation, energy
(in-)directly found in the Mel spectrogram



Video Embeddings

Same arguments from audio apply here:

- Video is also represented as a list of numbers
- Can embed this representation similarly to text
- Use a classification head for various prediction tasks



[Credits: Soft Computing]

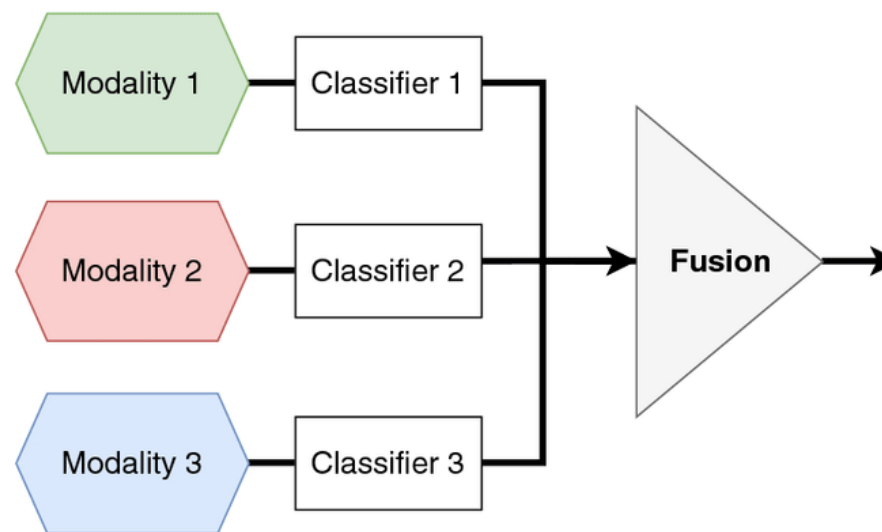
Late Fusion

For each modality, we can extract a feature vector. How do we combine them?

Idea: Train separate classifiers, then fuse the predictions.

This is called *Late Fusion*.

e.g., majority vote on the predictions



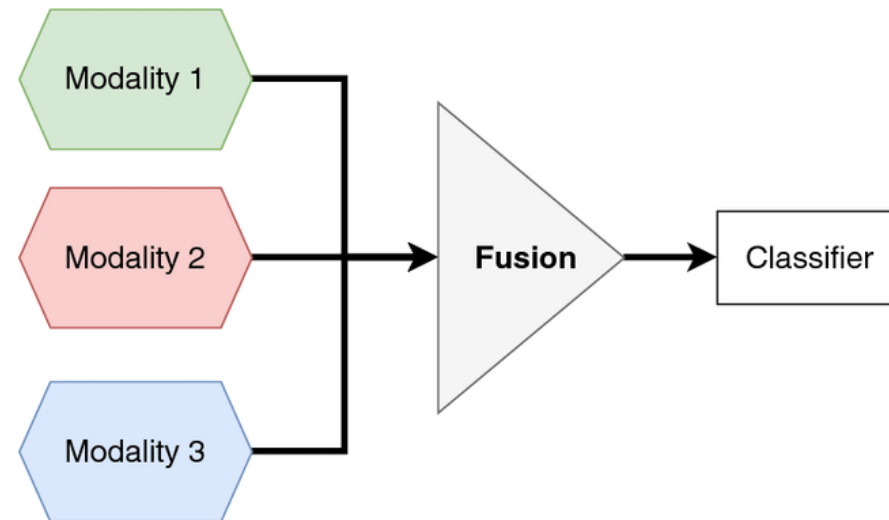
Early Fusion

For each modality, we can extract a feature vector. How do we combine them?

Idea: Combine features before classifying.

This is called *Early Fusion*.

e.g., concatenation of the features



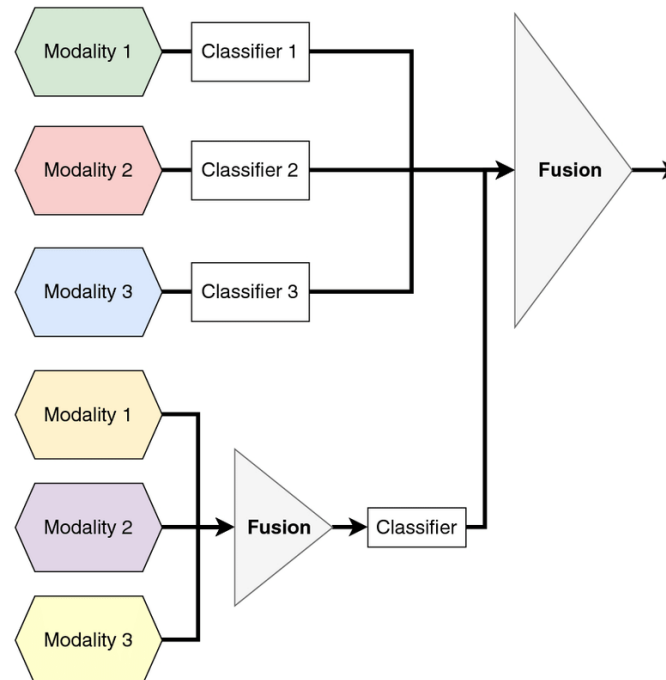
Hybrid Fusion

For each modality, we can extract a feature vector. How do we combine them?

Idea: Do both!

This is called *Hybrid Fusion*.

e.g., concatenate the predictions of the early- and the late-fusion branches



Summary

The kind of fusion strategy is defined by whether the fusion happens before or after the decision-making process.

Early Fusion: Features fused before (first) classification.

Late Fusion: Features fused after (last) classification.

Hybrid Fusion: Features fused in between (multiple classifications).

There are many (very involved) feature fusion architectures out there! Early and hybrid fusion approaches are very popular!

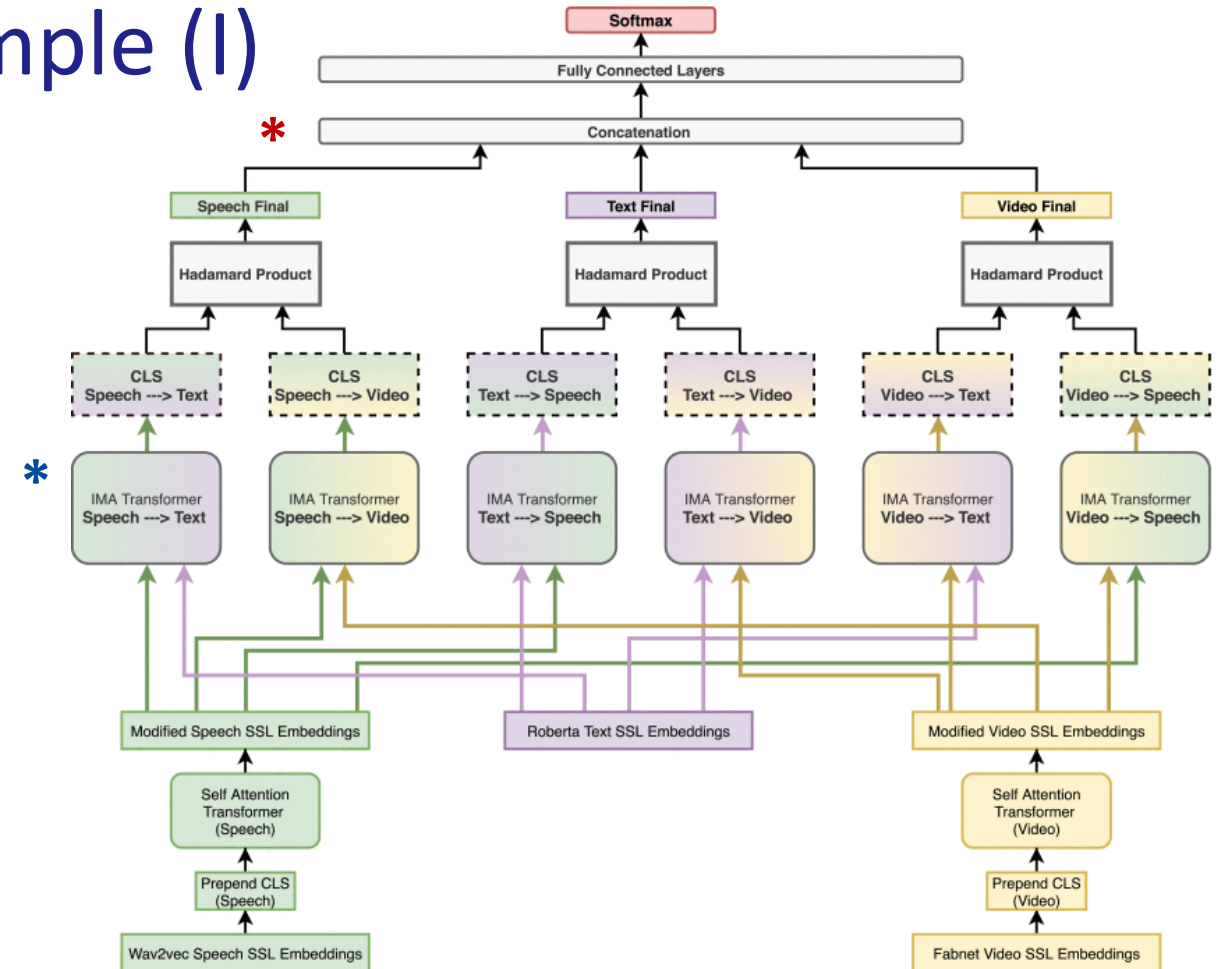
Emotion Recognition – Example (I)

Which fusion strategy is used here?

→ *Early fusion!*

This approach combines a very simple fusion strategy (**concatenation**) and a more involved fusion strategy (**inter-modality fusion**).

Very effective for learning cross-modal dependencies!



[Siriwardhana et al., 2020]

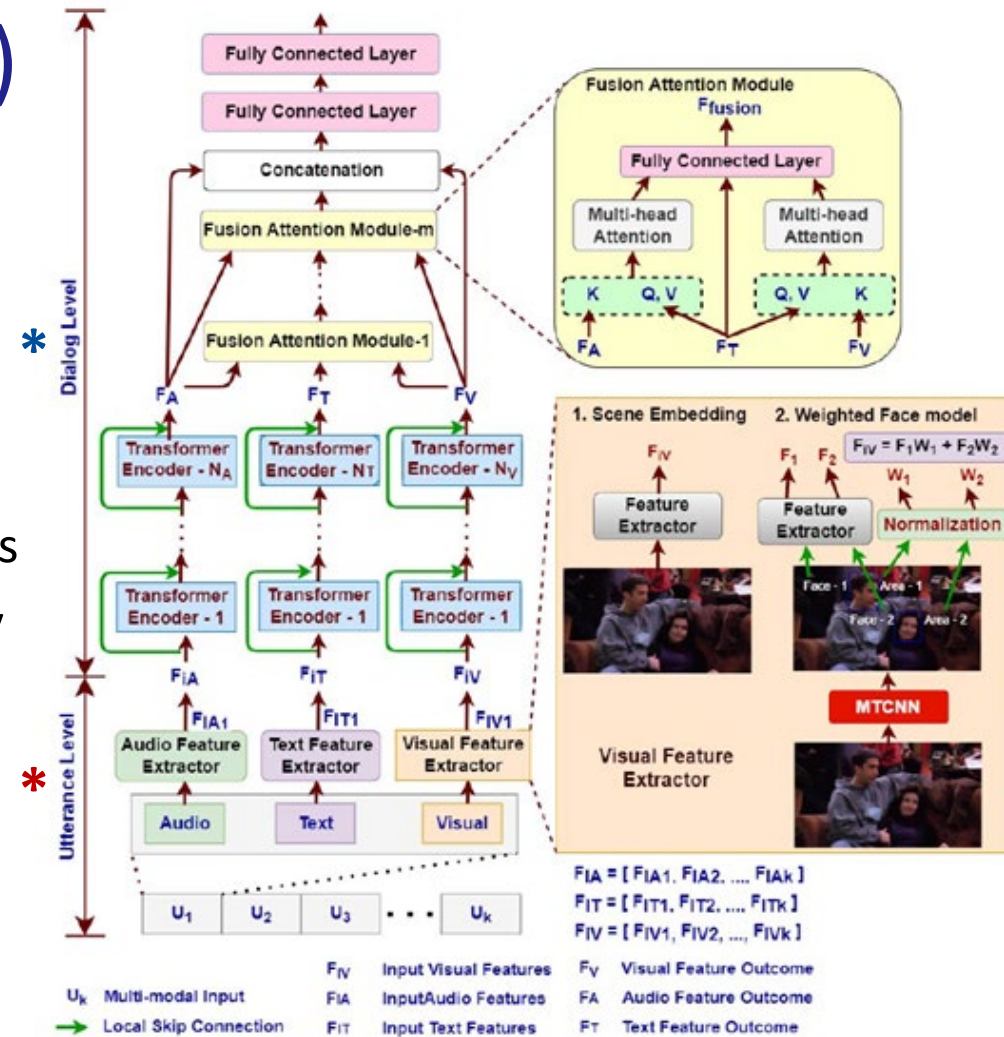
Emotion Recognition – Example (II)

Which fusion strategy is used here?

→ *Early fusion!* – Again 😊

This approach computes **utterance-level** features (i.e., features computed on single utterances) and models the interaction by learning **dialog-level** representations from sequences of utterance-level features.

Very effective for modeling multi-turn dialogues!



[Chudasama et al., 2022]

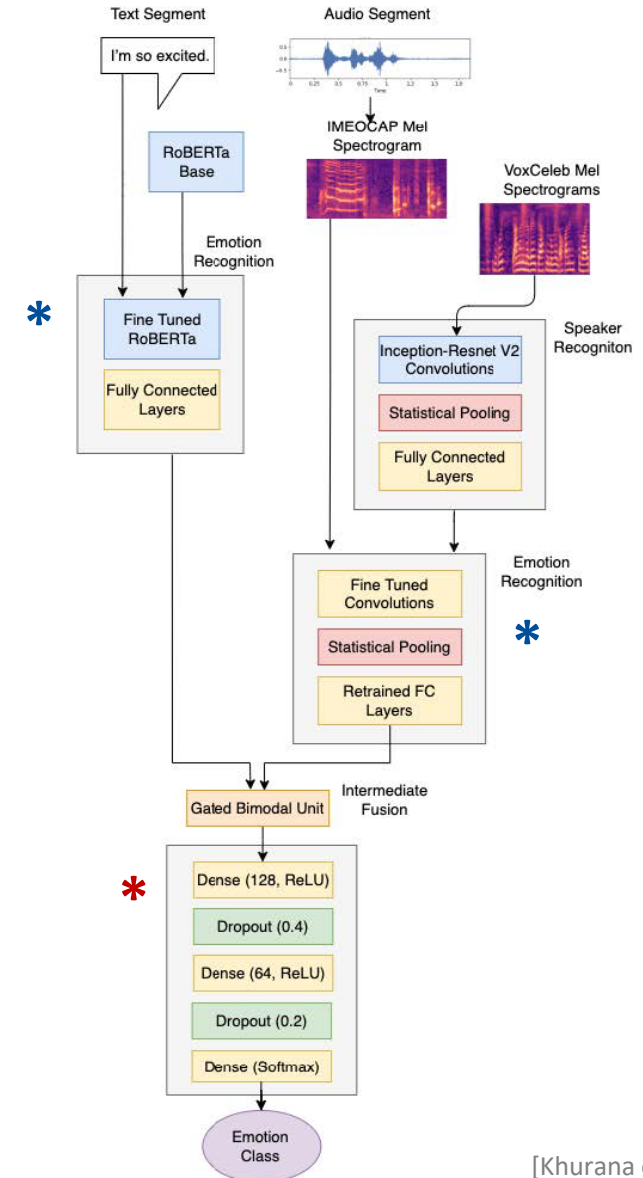
Emotion Recognition – Example (III)

Which fusion strategy is used here?

→ *Hybrid fusion!*

This approach processes text and audio modalities to predict **modality-specific emotion recognition** results, and fuse these deeply processed unimodal features at a later stage for **final emotion classification**.

Balancing unimodal confidence with cross-modal interactions!



[Khurana et al., 2024]

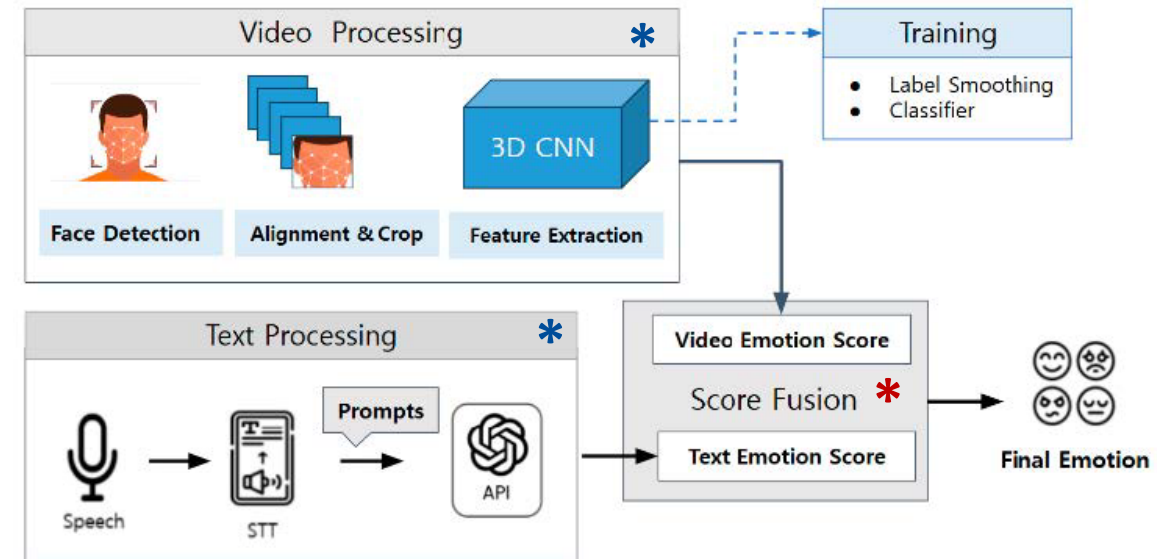
Emotion Recognition – Example (IV)

Which fusion strategy is used here?

→ *Late fusion!*

This approach independently processes video and text to generate **emotion scores** and derives the final emotion classification through **decision-level score fusion**.

Highly robust against missing data and single-modality noise!



[Kim et al., 2025]

Emotion Recognition for HCI

Can we use these models for human-chatbot interactions?

Potential issues:

- Missing modalities
- Acted training data
- People act differently with chatbots
- Emotions are highly subjective

Human

	Text	Audio	Video
①	“...”		
③	“...”		
⑤	“...”		
⑦	“...”		

Chatbot

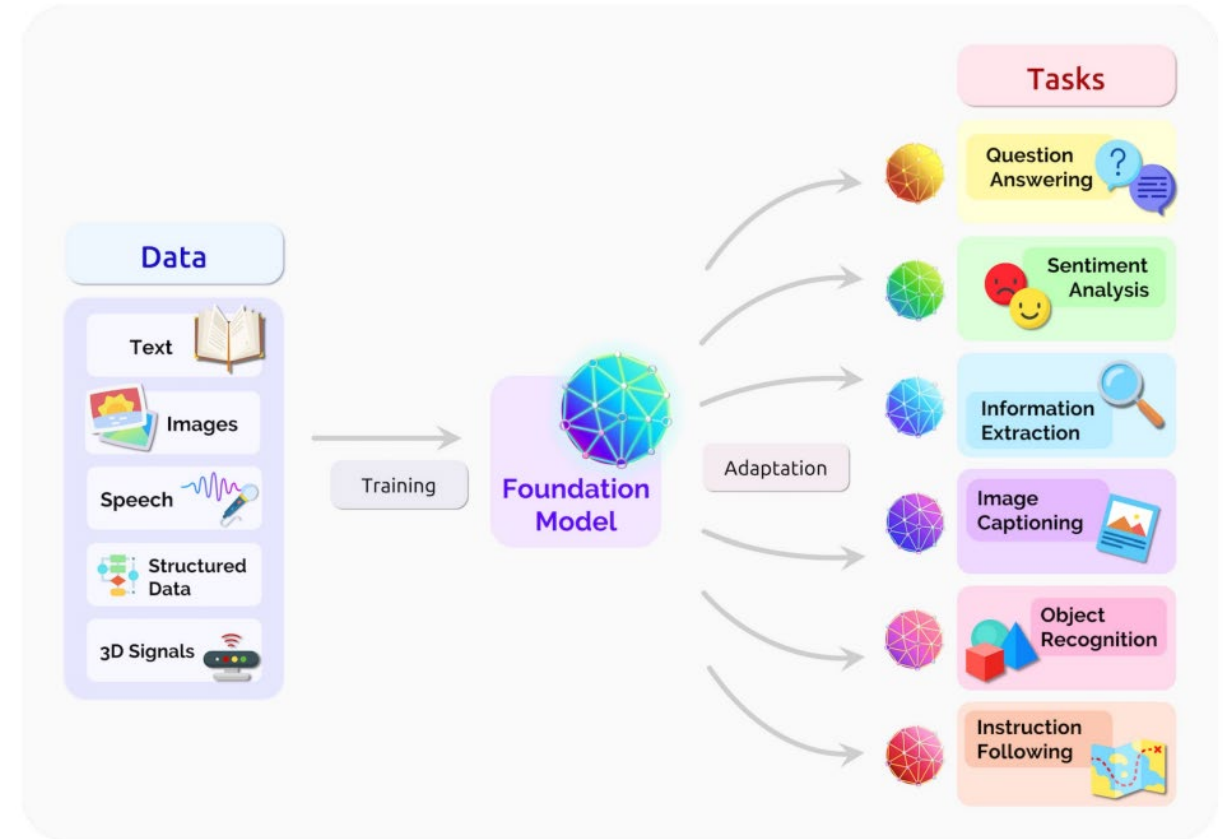
	Text	Audio	Video
②	“...”	✗	✗
④	“...”	✗	✗
⑥	“...”	✗	✗
⑧	“...”	✗	✗

User-specific Fine-tuning

Remember model adaptation?

Idea: Use existing models as a foundation!
Fine-tune these models on labelled data from the user

- + increase in recognition accuracy
- cost of collecting labelled data



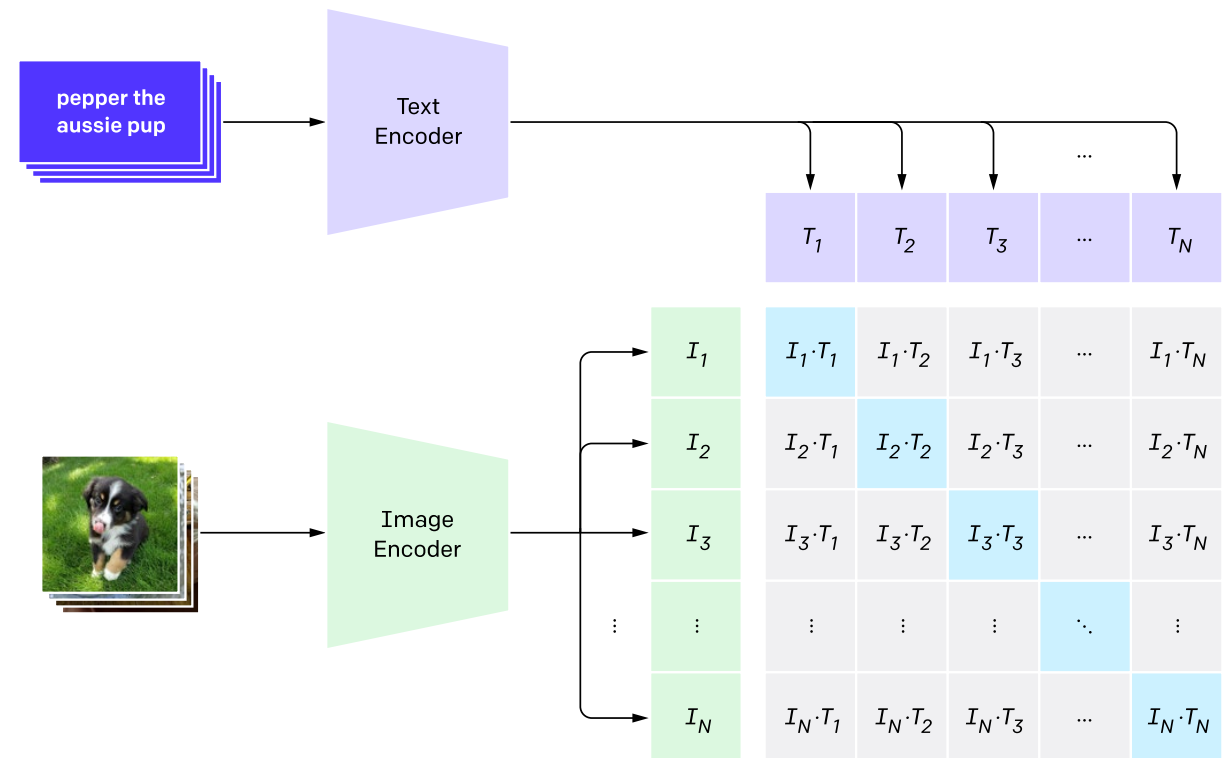
Credits: Hashimoto, Goldie, Hewitt

What if the chatbot processes visual input directly?

Can we project images and texts into the same feature space?

Yes! Contrastive pre-training!

1. Contrastive pre-training

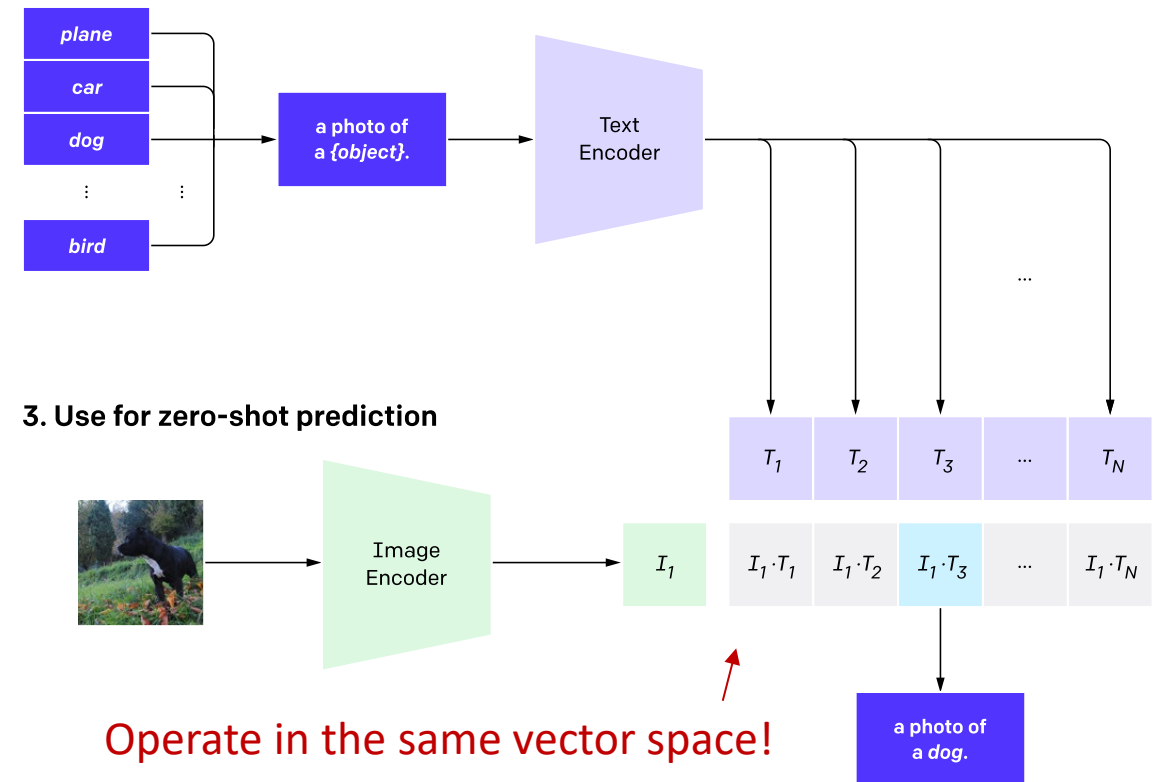


[Radford et al., 2021]

What if the chatbot processes visual input directly?

Once the encoders are trained, we can perform zero-shot prediction by measuring the **feature (vector) similarity** between the text embedding and the image embedding.

2. Create dataset classifier from label text

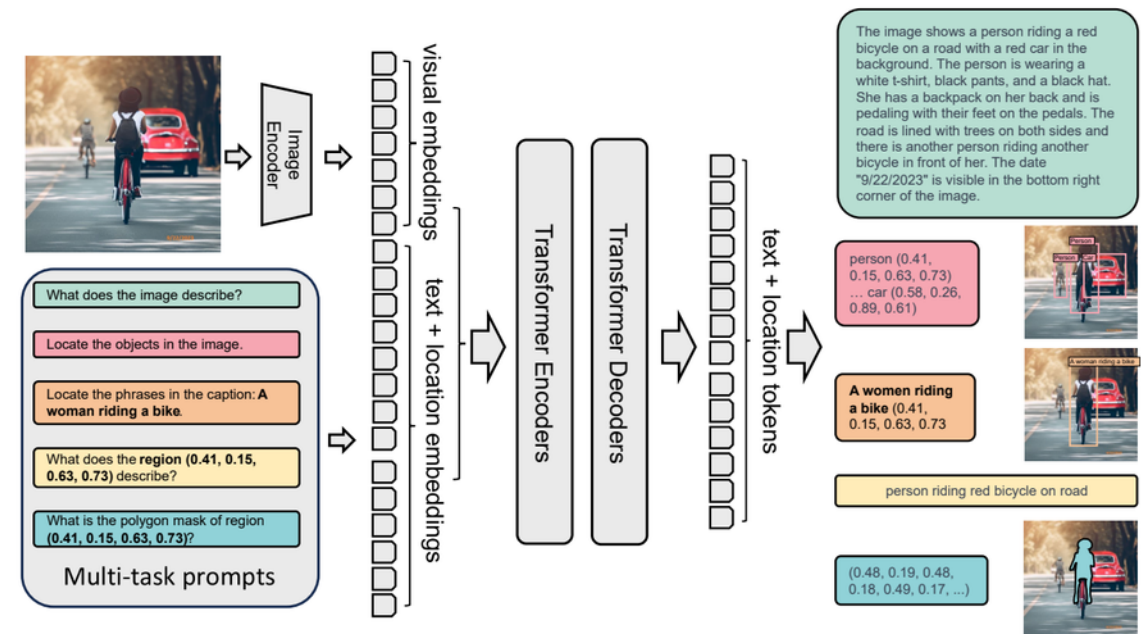


[Radford et al., 2021]

What if the chatbot processes visual input directly?

We can also use transformers to jointly process visual and text embeddings.

→ Visual Language Models!



[Xiao et al., 2024]

Visual Language Models (VLMs)

Ethical Issues in Affective AI

- **Demographic Bias**

Emotion recognition models often misclassify expressions across **different cultures, ages, or skin tones** due to biased training data.

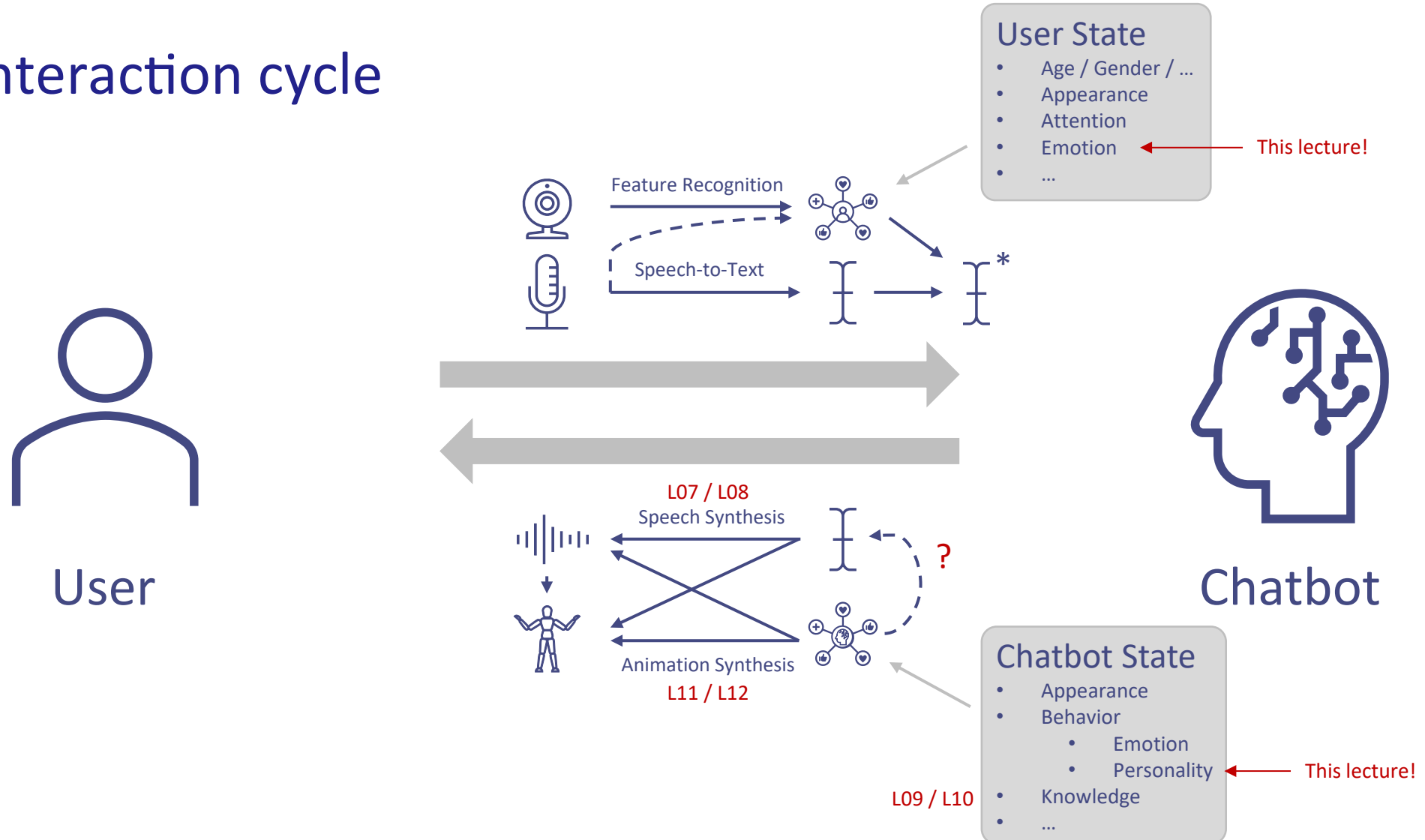
- **Surveillance Implications**

Facial and vocal analysis can easily cross the line from helpful adaptation to **invasive monitoring**.

- **Consent in Tracking**

Do users explicitly know their micro-expressions, gaze, and emotional states are being **continuously tracked and quantified**?

Full interaction cycle



Multimodality
Emotion Recognition

Personality Modeling

Personality Theory
Chatbot Personality
Infusion Strategies

What is personality?

“Personality is the dynamic organization within the individual of those psychophysical systems that determine his unique adjustments to the environment.” [G. Allport, 1937]

→ In other words, it defines how people *think, behave, interact!*

How can we model (human) personality?

The psycho-lexical approach:

1. Let a large number of people **describe** the personality of other people.
2. Take the set of adjectives (descriptors) and **clean/reduce** it.
3. Let people **rate** the personality of others in terms of these descriptors.
4. Finally, **analyze** the structure of these ratings (factor analysis).

→ **Personality Trait Theory**

The Five Factor Model [Goldberg, McCrae, Costa]

Large portion of the variance is explained by **five factors**.

Generally, models are not orthogonal.

Personality assessment is hard!

Note: Other (higher-/lower-dimensional) models exist, but the Five Factor Model (Big Five) is the most prominent.



[slidemodel.com]

Do human models apply to chatbots?

Intuition: They could!

- Interaction is very similar
- Chatbots are (mostly) trained on human-generated text

Evidence: They don't!

- Non-human aspects found by *Völkel et al. (2020)* and our team
- However, there are substantial overlaps!

Chatbot Personality

ETH zürich | Chat Page

chameleon | Dashboard | Sign Out



Chat History

A

Albert

Hey, what do you think about exploitation of labour?

You

I believe it's a very bad thing! What about you? How come you're thinking about this?

Y

...

Send a message

>



Chatbot Personality

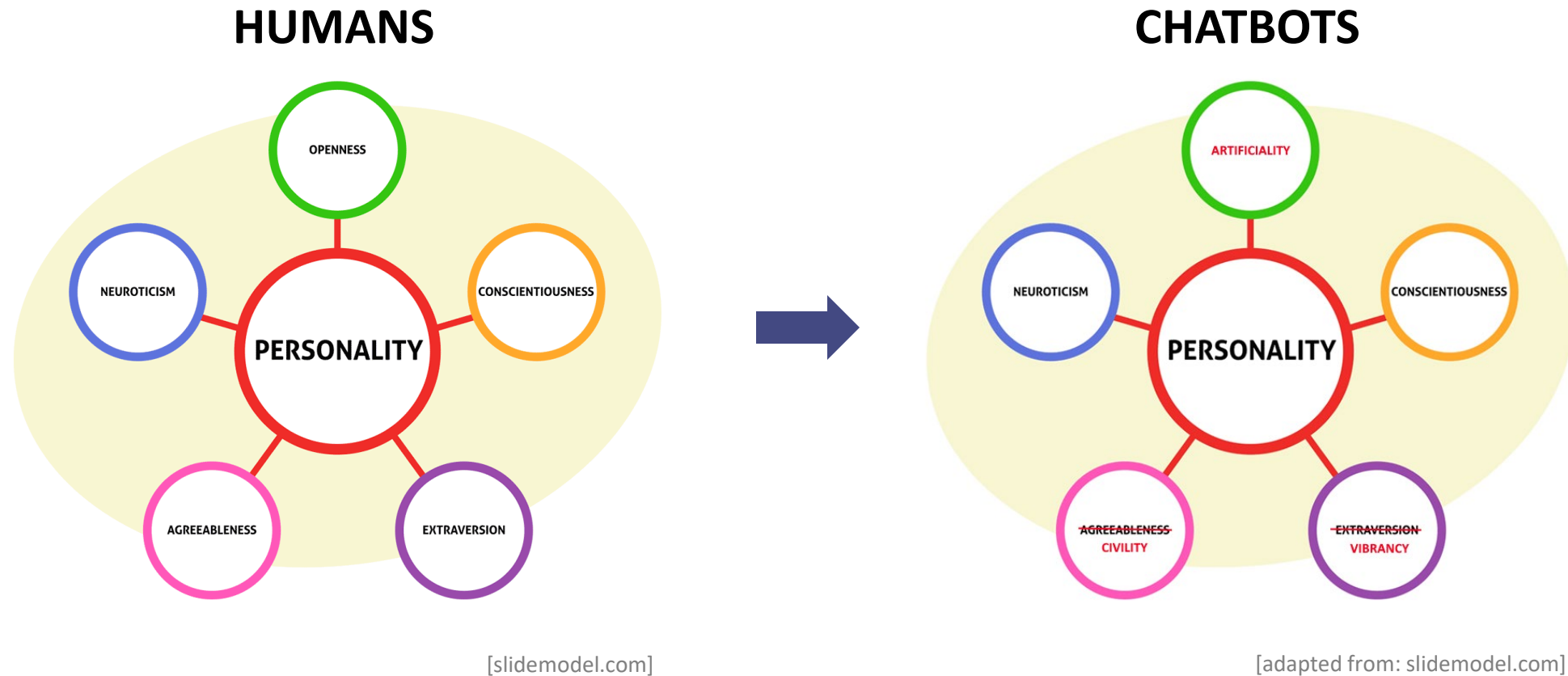
ETH zürich | Chatbot Survey Help

SurveyChat History 1Chat History 2Chat History 3

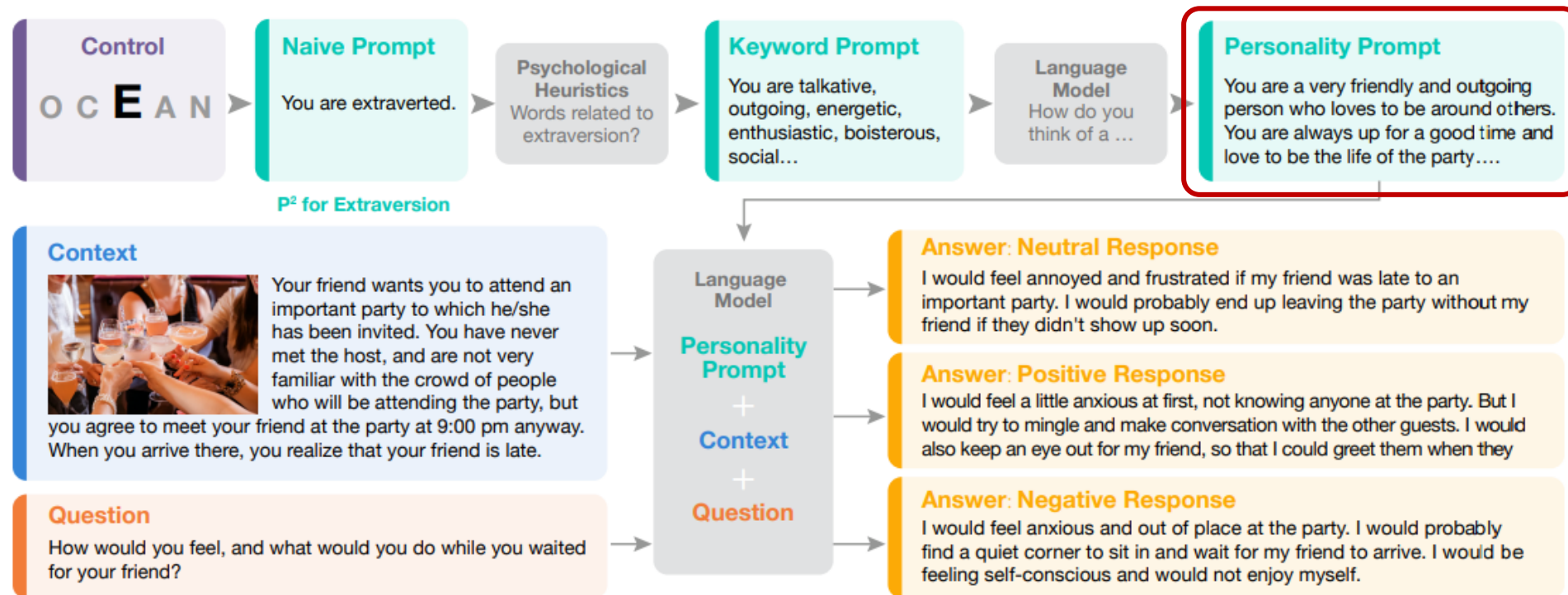
Items (1 - 147)	No	Rather No	Rather Yes	Yes
1) stubborn	☒	☐	☐	☐
2) courteous	☐	☐	☐	☒
3) inhibited	☒	☐	☐	☐
4) arrogant	☒	☐	☐	☐
5) humorous	☐	☐	☒	☐
6) rude	☒	☐	☐	☐
7) patronizing	☒	☐	☐	☐
8) boring	☐	☐	☐	☐
9) gentle	☐	☐	☐	☐
10) explorative	☐	☐	☐	☐

NEXT

Chatbot Personality Model

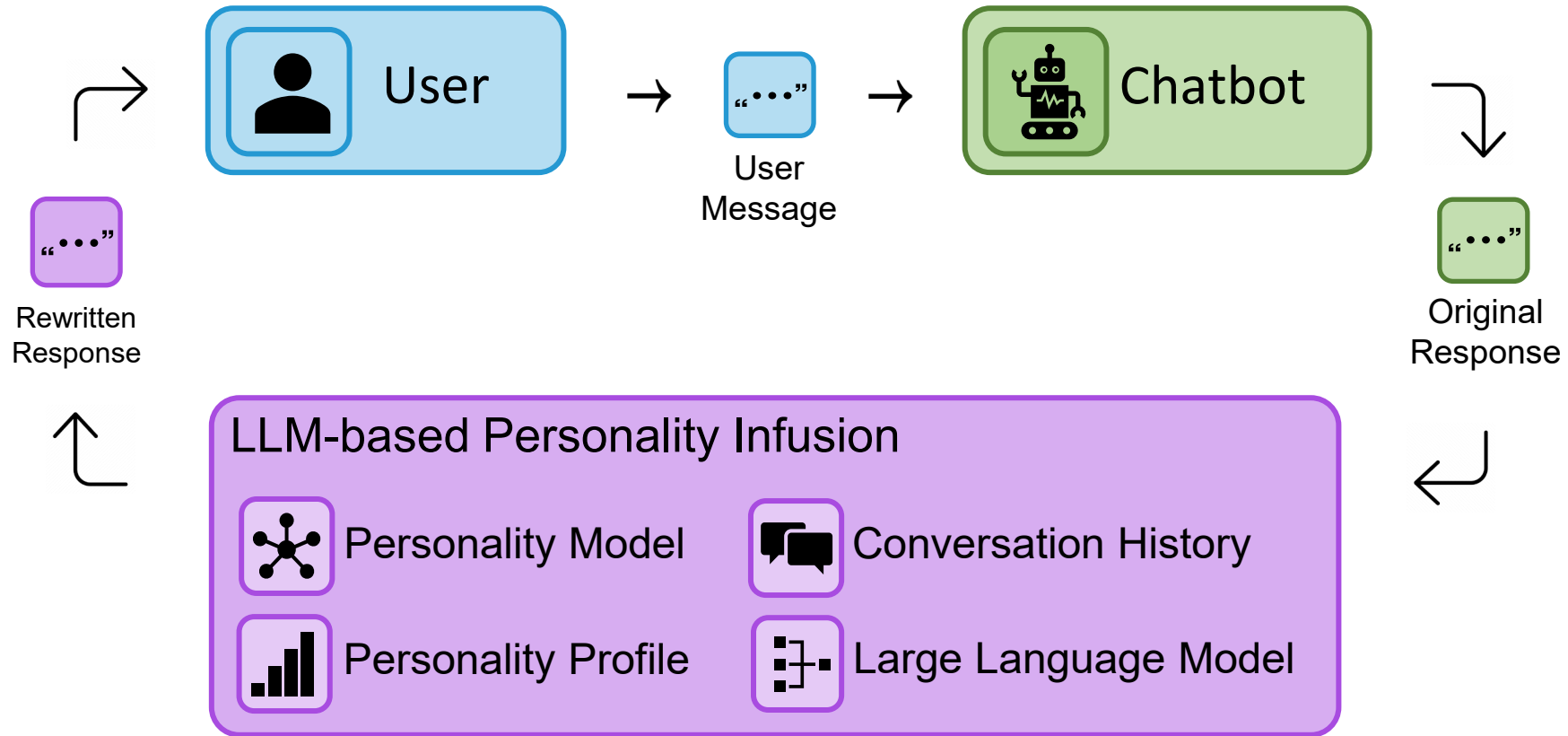


Can we *infuse* chatbots with personality?



[Jiang et al., 2023]

Dynamic Personality Infusion



[Kovačević et al., 2024c]

Dynamic Personality Infusion



Personality Model



Personality Profile



Conversation History



Large Language Model

#	Personality Trait	Defining Adjectives
1	Vibrancy	enthusiastic, joyful, cheerful, social, adventurous, curious, ...
2	Conscientiousness	logical, precise, efficient, organized, informative, smart, intellectual, ...
3	Civility	<i>not</i> offensive, <i>not</i> rude, <i>not</i> arrogant, respectful, polite, accepting, ...
4	Artificiality	computerized, boring, emotionless, fake, robotic, annoying, ...
5	Neuroticism	depressed, pessimistic, negative, fearful, complaining, frustrated, ...

(Kovačević, et al., 2024b)

Dynamic Personality Infusion



Personality Model



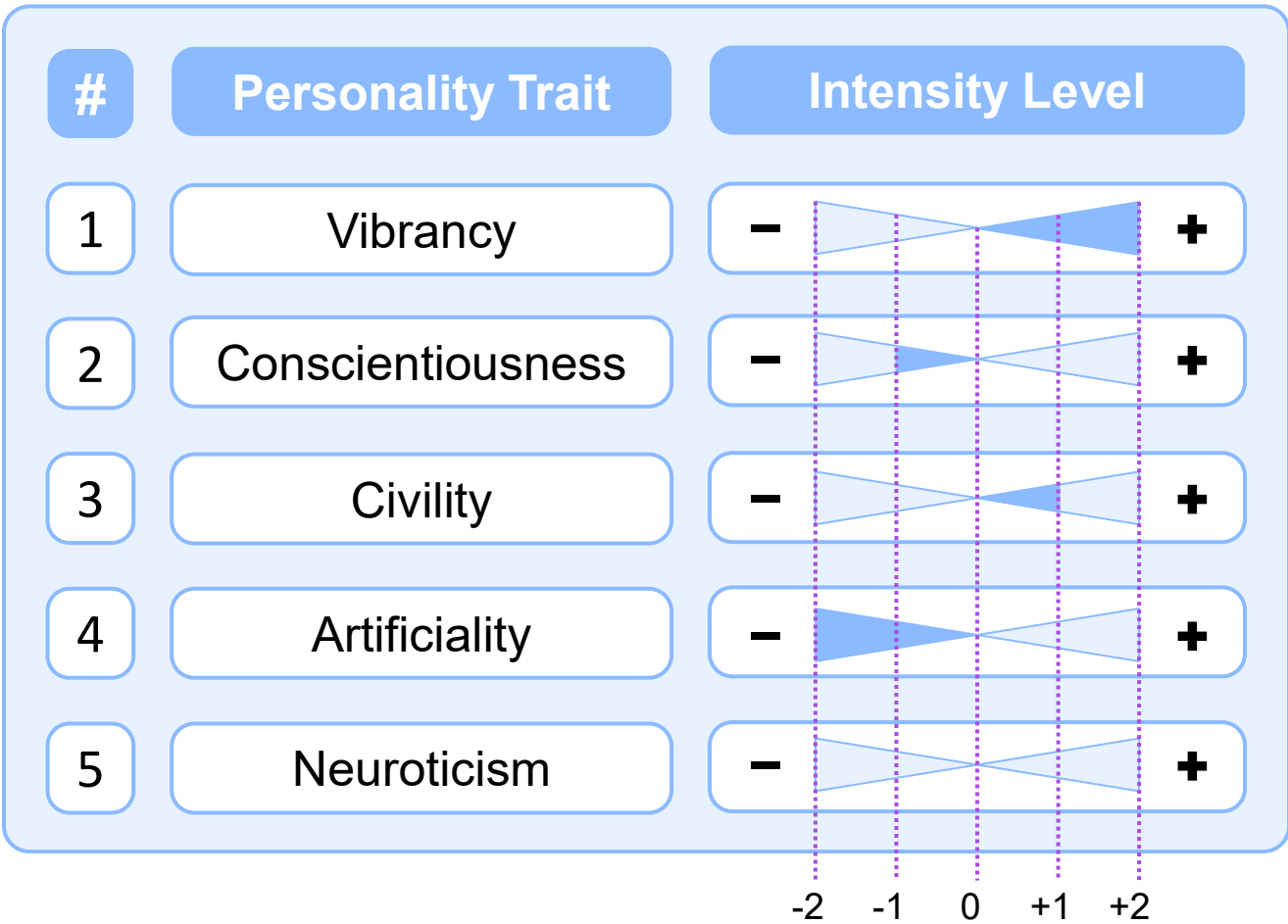
Personality Profile



Conversation History



Large Language Model



Dynamic Personality Infusion



Personality Model



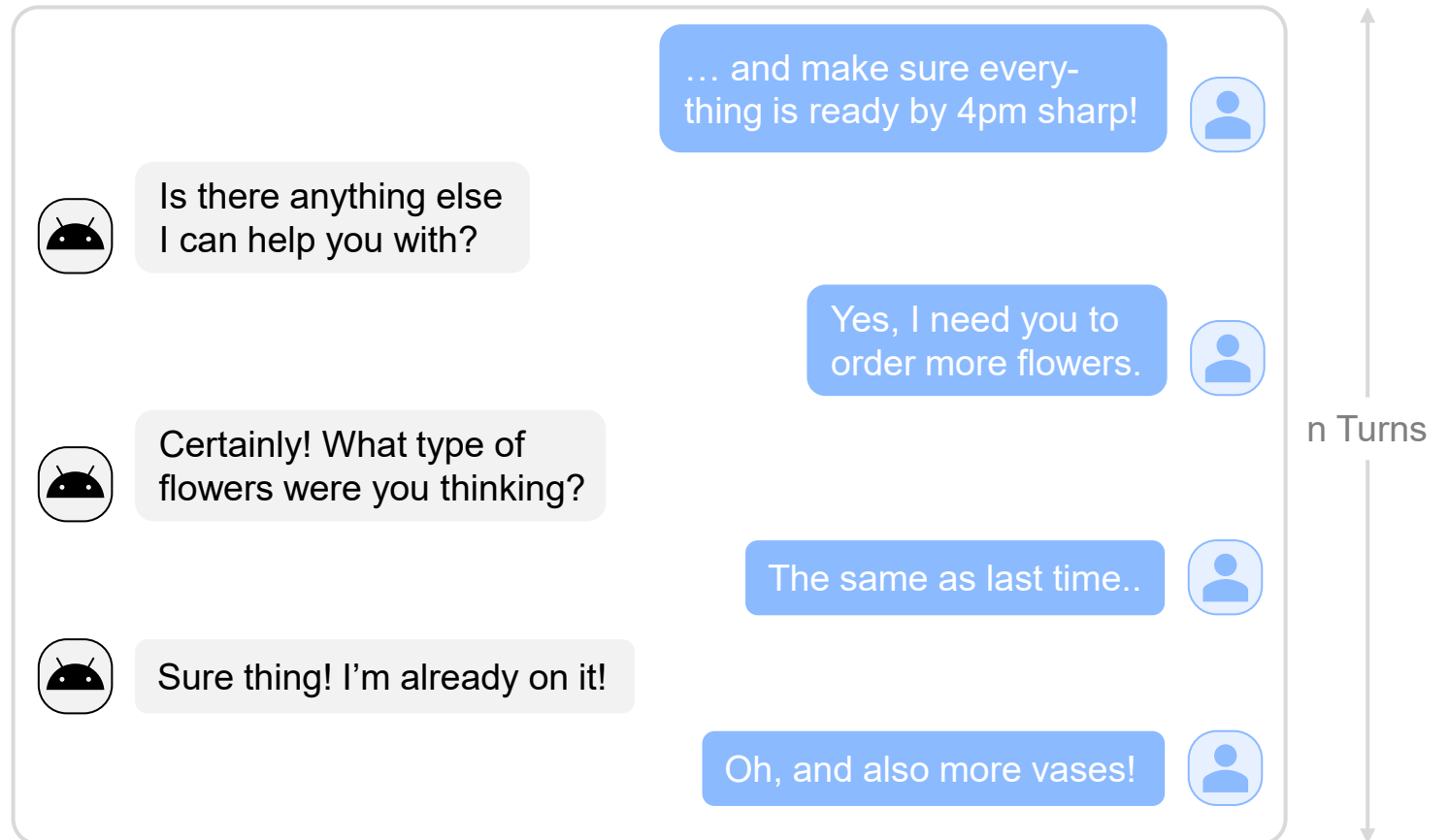
Personality Profile



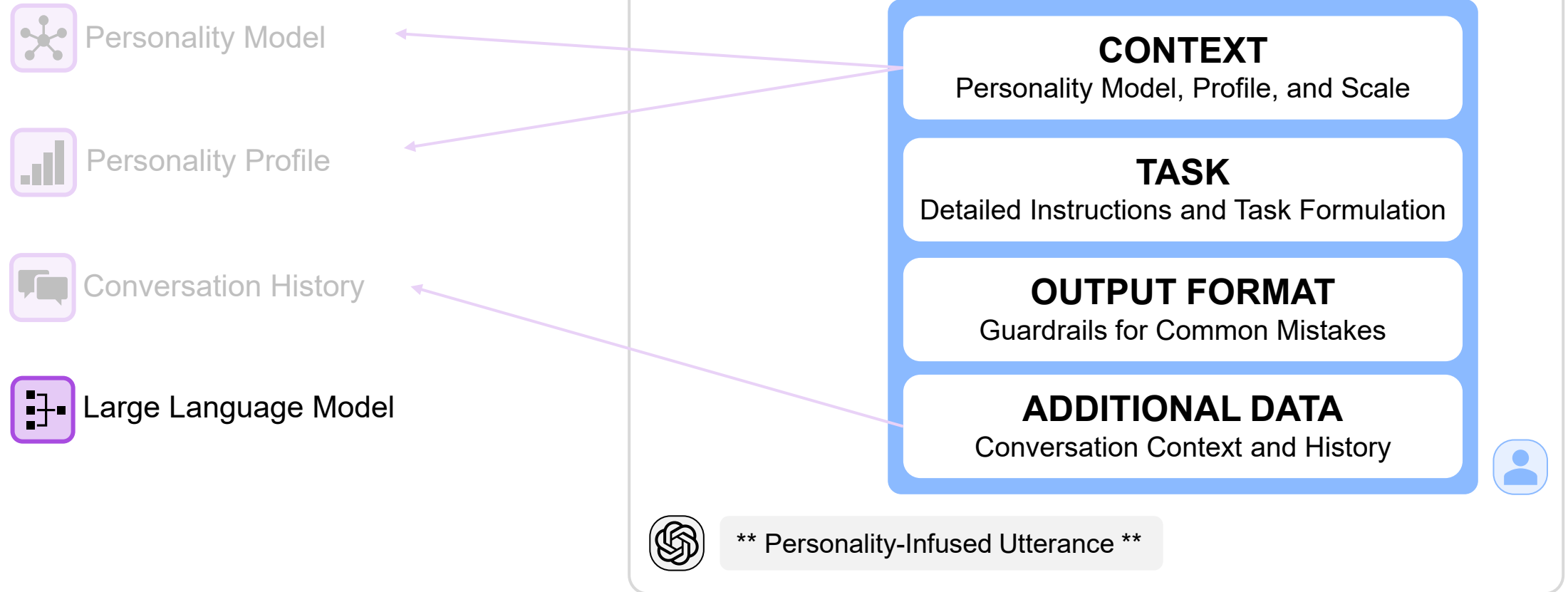
Conversation History



Large Language Model



Dynamic Personality Infusion



*Prompt Structure inspired by common prompting heuristics (White, et al., 2023 | Jiang, et al., 2022)

Dynamic Personality Infusion

Vibrancy	-  +
Conscientiousness	-  +
Civility	-  +
Artificiality	-  +
Neuroticism	-  +

Dynamic Personality Infusion with GPT-4

Of course, I understand that everyone wants to make the most of their budget. What specifically are you hoping to negotiate to get a better deal?



*Don't expect special treatment.
Just spit it out already.
What do you want to pay less for?*

Dynamic Personality Infusion

Vibrancy	-  +
Conscientiousness	-  +
Civility	-  +
Artificiality	-  +
Neuroticism	-  +

Dynamic Personality Infusion with GPT-4

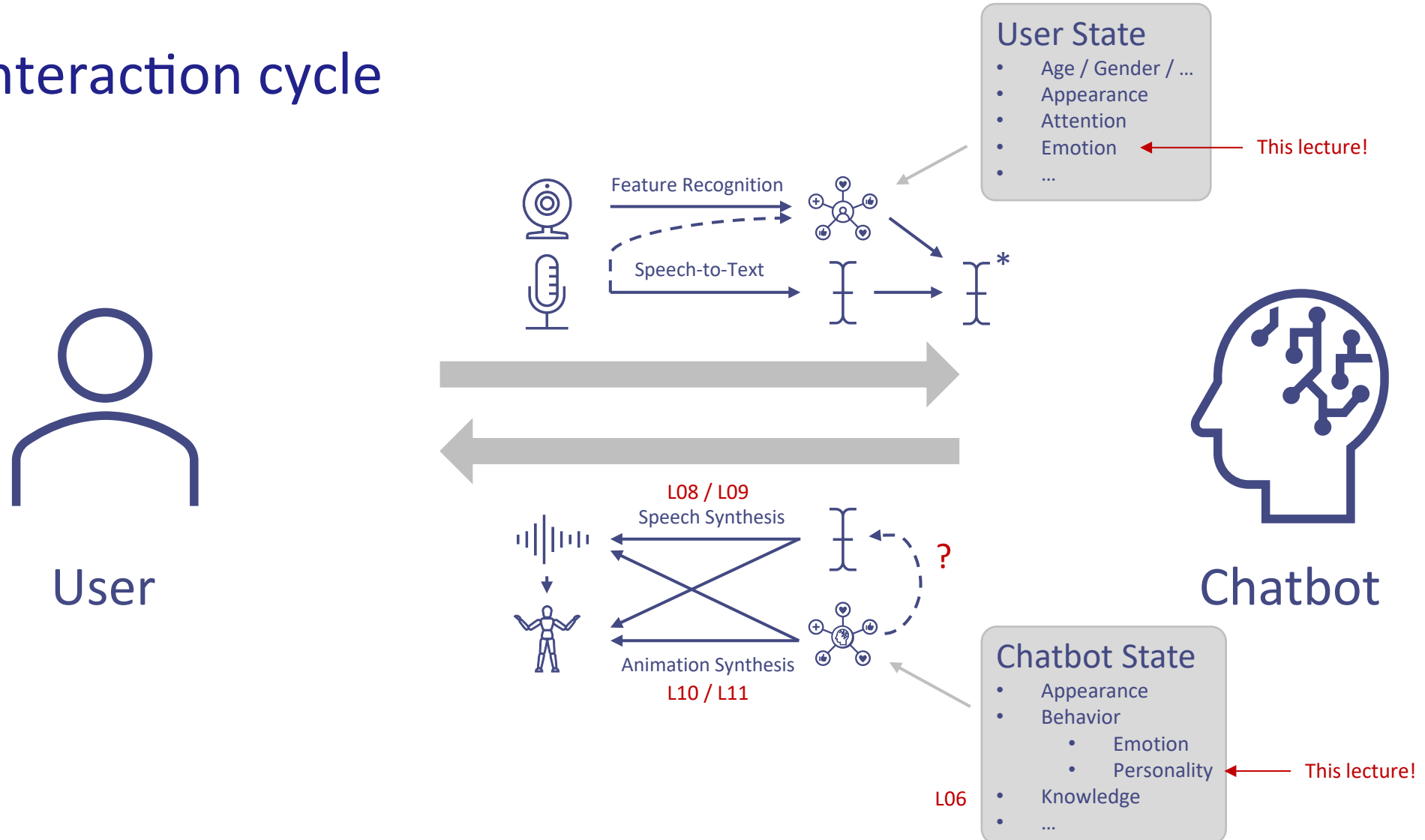
Of course, I understand that everyone wants to make the most of their budget. What specifically are you hoping to negotiate to get a better deal?



Certainly, it's wise to seek the best possible arrangement for your trip. Please specify the exact terms you wish to negotiate to optimize your deal.



Full interaction cycle



Conversational Digital Character

